

BASIC CONCEPT

1. Atomic weight of Ca is
A) 20 B) 40 C) 45 D) 80
2. One 12^{th} the mass of C_{12} is called
A) Atomic Mass B) A.m.u C) Both a and b D) None
3. Any substance having chemically identical atoms is called
A) Molecule B) Element C) Compound D) None
4. The Avogadro's number is:
A) 6.07×10^{-23} B) 6.022×10^{23} C) 6.022×10^{-23} D) 6.002×10^{-23}
5. The molecular formula and empirical formula may be
A) Identical B) Different C) Both a and b D) None
6. The empirical formula of chloroform is
A) CCl_4 B) CHCl_3 C) CH_2Cl_2 D) CH_2O
7. H_2O_2 is the chemical formula of
A) Water B) Methane C) Alcohol D) Hydrogen Peroxide
8. CH_4 is the chemical formula of
A) Ethane B) Propane C) Hydrogen D) Methane
9. The S.I unit for energy are
A) J B) Calorie C) K D) Cd
10. The standard for atomic mass is
A) H B) He C) Ne D) C
11. The amount of matter in substance is called its.
A) Weight B) Mass C) Volume D) Density
12. The C.G.S. units of energy is
A) Dynes/cm B) NM C) Kgm D) Dynes cm
13. The empirical formula of sugar is
A) CH B) CH_2O C) C_6H_6 D) CHO
14. Which is not branch of Chemistry?
A) Physical Chemistry B) Nuclear Chemistry
C) Electrical Chemistry D) Bio Chemistry
15. Combustion analysis is performed for determining
A) Number of Ions B) Empirical formula of organic compound
C) Structure Formula D) Isotopes of an element
16. 0.5 mole of CH_4 and 0.5 mole of SO_2 gases have equal
A) Volume B) Mass is grams
C) Total number of atoms D) Isotopes of molecules
17. What is the mass of water formed when 4 grams H and 64 grams of O_2 combined together
A) 66 grams B) 18 grams C) 36 grams D) 72 grams
18. The total number of covalent bond is 4.5 grams of water is
A) 6.02×10^{23} B) 6.02×10^{22} C) 3.01×10^{22} D) 3.01×10^{23}
19. What is the volume in cm^3 of 3.01×10^{23} molecules of O_2 gas at S.T.P.
A) 1000cm^3 B) 11000cm^3 C) 1120cm^3 D) 11200cm^3
20. 2.38 grams of uranium contains U-atoms.
A) 6.02×10^{22} B) 6.02×10^{21} C) 3.01×10^{22} D) 3.01×10^{21}
21. A compound having empirical formula CH_3O and its molecular mass is 110.02 its molecular formula is
A) $\text{C}_3\text{H}_3\text{O}$ B) $\text{C}_6\text{H}_6\text{O}_6$ C) $\text{C}_9\text{H}_9\text{O}_3$ D) $\text{C}_3\text{H}_6\text{O}_2$
22. On dividing percentage of elements by their atomic masses we get.
A) Empirical Formula B) Atomic ratio of elements

C) Mole ratio of elements

D) Molecular formula of compound

23. In the determination of atomic ratio of the element the mole ratios are divided by
A) Least value of gram atoms of elements B) Atomic masses of elements
C) Given mass of the compound D) Molecular mass of the compound
24. Which one of the following steps is not involved in determination of empirical formula?
A) Determination % of each element
B) Determination of gram atoms of each elements
C) Determination of isotopes of each elements
D) Determination of atomic ratio of elements
25. Which one of the following statement is incorrect
A) Actual yield is always less than the theoretical yield
B) The formula of compound is not definite
C) Law of conservation of mass is applied in stoichiometry
D) Law of definite proportions applied in stoichiometry
26. Which of the following salts will have greater positive charge in its molar aqueous solution?
A) $\text{Ca}(\text{Cl})_2$ B) KCl C) NH_4Cl D) NaCl
27. A molecule is the smallest particle of a substance because
A) It has positive charge on it. B) It exists independently
C) It decomposes into ions D) It is always mono atomic
28. Molecular formula is determined from empirical formula because
A) Molecular formula $\times$ empirical formula = n B) Molecular formula = n $\times$ empirical formula
C) Molecular formula $\times$ n = empirical formula D) Molecular mass empirical formula = n
29. A beaker contains 9 grams of water. The number of H atoms is
A) 6.02×10^{23} B) 3.01×10^{23} C) 6.02×10^{24} D) 3.01×10^{-23}
30. One mole of $\text{C}_2\text{H}_5\text{OH}$ contains no. of oxygen atoms
A) 6.02×10^{23} B) 3.61×10^{24} C) 1.81×10^{24} D) 6.02×10^{24}
31. 9.8 grams of aqueous solution of H_2SO_4 contains moles of H^+ ions.
A) 0.1 B) 0.2 C) 0.1 D) 0.01
32. When 0.1 g of Mg is treated with an excess of HCl, what volume of gas at STP will be produced?
A) 10cm^3 B) 24cm^3 C) 48cm^3 D) 93cm^3
33. A compound contains 83% C & 17% H. Empirical formula of the compound is _____.
A) C_2H_2 B) C_2H_4 C) C_2H_5 D) C_2H_6
34. 8g of CH_4 on complete combustion gives 18g CO_2 . The % yield is _____.
A) 99.2% B) 95.5% C) 86.27% D) 81.81%
35. _____ has greater number of moles.
A) 0.1g Na B) 6.02×10^{20} atoms of Mg
C) 0.1 mol/dm³ NaOH in 20 cm³ D) 12.2 dm³ N_2 at STP
36. The specific gravity of H_2SO_4 is 1.84, its density would be _____.
A) 0.0184 g/ml B) 0.184 g/ml C) 1.84 g/ml D) 18.4 g/ml
37. X^{+1} ion contains 18 electrons and 20 neutrons, what is mass number?
A) 39 B) 37 C) 38 D) 20

GASES

1. The collision between gas molecules are
A) Elastic B) Inelastic C) Both A and B D) None
2. Different gases with same K.E. move with _____ velocities.
A) Same B) Different C) Equal D) None
3. Intermolecular forces are _____ than binding forces.
A) Stronger B) Equal C) Weaker D) None
4. The rate of diffusion of light gases is _____ than heavier gases.
A) More B) Less C) Equal D) None
5. What is ratio of diffusion NH_3 and HCl ?
A) 1.46 B) 1.8 C) 2.0 D) 1.2
6. If $V_1=5$ liters, $P_1 = 2$ atm, $T_1 = 273^\circ\text{C}$. $T_2 = 0^\circ\text{C}$ and $V_2 = ?$ When $P_2=1$ atm.
A) 5 lit B) 10 lit C) 2.5 lit D) 12.5 lit
7. $\text{K.E.} \propto T$ is expression for
A) Boyle's Law B) Charles's Law C) Kinetic Molecular Theory D) None
8. What is the ratio of rates diffusion of equal volume of SO_2 and CH_4 ?
A) 2 B) 1.6 C) 1.0 D) 2.1
9. Absolute zero is equal to
A) 0°C B) -459.7°F C) -273°K D) None
10. Barometer is used to determine
A) Pressure B) Absolute Temperature C) K.E. D) Length
11. Equal volumes of all gases at STP contain equal no of molecules is called?
A) Dalton's Law of Partial Pressure B) Graham's Law of Diffusion
C) Avogadro's Law D) None
12. The spontaneous homogenous mixing of molecules of different gases by random motion and collision called
A) Diffusion B) Effusion C) Dalton's Law D) None
13. Mathematical expression for Charles's Law is
A) $V \propto T$ B) $V/T = \text{Constant}$ C) Both A and B D) $V \propto 1/t$
14. $V \propto 1/P$; is mathematical expression for
A) Charles's Law B) Avogadro's Law C) Boyle's Law D) Brown's Law
15. S.I units for measurements of pressure.
A) Pascal B) mm of Hg C) atm. D) Torr
16. Absolute zero is equal to
A) 273°C B) -273°C C) -273°K D) -273°F
17. Which is correct?
A) 1 mm Hg = 1 torr = 1 atm B) 1 mm Hg = 760 torr = 760 torr = 1 atm
C) 760 mm Hg = 760 torr = 1 atm D) 760 mm Hg = 1 torr = 1 atm
18. The gas law are not obeyed by general gases at
A) Low Temp B) High Pressure C) High Temp D) Both A and B
19. For a gas the isotherm is the graph between
A) V B) P C) Both A and B D) None
20. One atmosphere is equal _____ Pascal
A) 760 B) 101325 C) 14.7 D) 1.01325
21. Which is not the state of matter?
A) Plasma B) Solid C) Gas D) None
22. Weight of one dm^3 of O_2 at STP is
A) 1.4384 B) 2.531 C) 1.6384 D) 1.8384
23. All gases liquefy before reaching at
A) 273°K B) 373°K C) 0°K D) 73°C

24. Most ideal gas at room temperature is
 A) CO_2 B) NH_3 C) SO_2 D) N_2
25. Absolute temperature of a gas is proportional to
 A) Translational Kinetic Energy B) Rotational Kinetic Energy
 C) Vibrational Kinetic Energy D) Potential Energy
26. Density of a gas is usually expressed in these chemical units:
 A) Kg m^3 B) Kg m^3 C) g dm^{-3} D) g cm^{-3}
27. Which of the following has same number of molecules at STP
 A) 1000 cm^3 of N_2H_4 and O_2 B) 200 cm^3 of CO_2 and N_2O
 C) 50 cm^3 each of CO and N_2 D) All above
28. The deviation of a gas from ideal behavior is maximum at
 A) -10°C and 5.0 atm B) -10°C and 2.0 atm
 C) 100°C and 2.0 atm D) 0°C and 2.0 atm
29. The molar volume of CO_2 is maximum at:
 A) STP B) 127°C and 1 atm C) 0°C and 1 atm D) 273°C and 1 atm
30. If absolute temperature of gas is double and the pressure is reduced to half the volume of gas will.
 A) Remain Unchanged B) Increases Four Times
 C) Reduce to $1/4^{\text{th}}$ D) Be Doubled
31. Pressure remaining constant at which temperature the volume of gas will become twice of what it is at 0°C .
 A) 545°C B) 200°C C) 546 K D) None Of These
32. The order of the rate of diffusion of gases NH_3 SO_2 Cl_2 and CO_2 is
 A) $\text{NH}_3 > \text{SO}_2 > \text{Cl}_2 > \text{CO}_2$ B) $\text{NH}_3 > \text{CO}_2 > \text{SO}_2 > \text{Cl}_2$
 C) $\text{Cl}_2 > \text{SO}_2 > \text{CO}_2 > \text{NH}_3$ D) None of them
33. Helium gas diffuses twice as fast as another gas B. If the molecular weight of Helium is 4, the molecular weight of gas B is _____.
 A) 4 B) 8 C) 16 D) 24
34. 1 torr is equal to _____.
 A) 2 mm Hg B) 1 mm Hg C) 100 mm Hg D) 1000 mm Hg
35. A sample of a gas contains 300ml at 27°C and 1 atm. Pressure. Calculate the volume of the gas at 32°C and 5 atm. Pressure.
 A) 110 ml B) 0.05 L C) 0.061 dm^3 D) 71 cm^3
36. A real gas is expected to behave non-ideally at _____.
 A) $\uparrow T, \downarrow P$ B) $\downarrow T, \downarrow P$ C) $\downarrow T, \uparrow P$ D) $\uparrow T, \uparrow P$
37. At constant temperature the volume of gas is 95 cm^3 at a pressure of $9.961 \times 10^4 \text{ N/m}^2$. What will be its volume at $10.13 \times 10^4 \text{ N/m}^2$?
 A) 1 cm^3 B) 97 cm^3 C) 93 cm^3 D) 10 cm^3

ATOMIC STRUCTURE

- If $l = 2$ then the orbital will be _____ orbital.
A) s B) p C) d D) f
- Pauli's principle is applicable to.
A) Degenerate Orbitals B) Two electrons in the same orbital
C) One electron D) None
- Magnetic quantum number represents _____ of orbitals.
A) Size B) Shape C) Orientations D) Nuclear Stability
- Balmer's series is in _____ region.
A) Visible B) U C) I.R. D) None
- Which is the units of length
A) Angstrom B) Poise C) Equal D) None
- Photons of yellow color are _____ energetic than violet colour
A) More B) Less C) Equal D) None
- The instrument which measures the absorbance or emission of intensity and frequency is called
A) Spectrophotometer B) Perimeter
C) Glucometer D) Calorimeter
- The red colour have the wavelength _____ nm.
A) 600 – 300 B) 630 – 750 C) 100 – 200 D) None
- Which have better penetrating power.
A) Alpha Rays B) Beta Rays C) Gamma Rays D) X-Rays
- Alpha rays are actually
A) 1 Proton 2 Neutrons B) 2 Proton 3 Electrons
C) 2 Proton 3 Neutrons D) 2 Proton 1 Neutrons
- The lightest and the simple canal rays are formed in gas discharge tube when _____ is filled.
A) N B) He C) H D) O
- The charge value of an electron is determined by
A) J. J. Thomson B) Crooks C) Perrin D) R. A. Millikan
- Which is lighter than the rest
A) Electron B) Proton C) Neutron D) None
- 2S orbital is _____ in energy than 1S.
A) Higher B) Lower C) Equal D) None
- Spin quantum number may be
A) 0 B) 1 C) 2 D) None
- $E = h \nu$ is the
A) Spectral equation B) Plank's Equation
C) De Brogie Equation D) Heisenberg's Equation
- The emission of radiations form nuclei of light atoms by the bombardment of the moving particles is called
A) Radioactivity B) Artificial Radioactivity
C) X-Ray Production D) X-Rays
- Their e/m , ratio resembles with that of electrons
A) Alpha Rays B) Beta Rays C) Gamma Rays D) X-Rays
- The value of R (Rydberg's Constant) is _____ m^{-1} .
A) 1.0974×10^7 B) 1.0842×10^7
C) 1.082×10^7 D) Both A and B
- The particle detector used for the discovery of the nucleus is composed of
A) Metal Foil B) ZnS C) Al D) Na

21. $\Delta X \cdot \Delta P = \frac{h}{4\pi}$ is equation is called
 A) Plank's Equation B) De Broglie's Equation
 C) Heisenberg's Equation D) None
22. The mass of electron is
 A) 1.602×10^{-10} Kg B) 1.6022×10^{-19} C) 9.1109×10^{-31} D) None
23. A spinning electron creates
 A) Magnetic Field B) Electric Field C) Zero D) None
24. The uncertainly principle was stated by
 A) De Broglie B) Heisenberg C) Einstein D) Schrodinger
25. $n + l$ value for 4f will
 A) 2 B) 5 C) 7 D) 9
26. Maximum number at electron in M-Shell
 A) -2 B) 8 C) 18 D) 32
27. The degenerate orbitals p-sub shell are.
 A) 2 B) 3 C) 5 D) 7
28. A dual character of matter particles in motion was postulated by
 A) De-Broglie B) Planck C) Einstein D) Schrodinger
29. The value of Plank's Constant 'h' is
 A) 6.625×10^{-34} B) 6.625×10^{-34} J sec C) 6.625×10^{-34} KJ D) 6.625×10^{-34} K Cal
30. Goldstein discovered that besides the cathode rays, another types of rays are produced in the discharge tube which are called
 A) X -Rays B) Beta Rays C) Positive Rays D) Gamma Rays
31. When the pressure of the gas in discharge tube is reduced, which of the following become more prominent
 A) Gas Glows B) Gas Ionizes
 C) A discharge Takes Place D) Gas Conducts Electricity
32. The radius of first orbit of hydrogen atom
 A) $0.329 \text{ \AA}$ B) $0.429 \text{ \AA}$ C) $0.529 \text{ \AA}$ D) $0.229 \text{ \AA}$
33. When the 6d orbital is completed the entering electron goes into
 A) 7f B) 7s C) 7p D) 7d
34. In the ground state of an atom the electron is present
 A) In the nucleus B) In the second shell
 C) Nearest to the nucleus D) Farthest from the nucleus
35. Cathode ray experiment confirms the present of _____
 A) Proton B) Electron C) Neutron D) None
36. The element having the lowest IP value has _____
 A) Large Atomic Size B) Higher Shielding Effect
 C) Small Nuclear Charge D) All
37. Line spectrum is obtained due to _____ of an electron.
 A) Atoms B) Ions C) Molecules D) All
38. Gamma rays have _____.
 A) Unit Mass & Negative Charge B) Zero Mass & Zero Charge
 C) Zero Mass & Positive Charge D) All
39. Maximum possible no of electrons in an energy level or shell 'n' is _____.
 A) $2n^2$ B) $4n$ C) $4n^2$ D) $2n$
40. The highest energy level which an electron can have is _____ levels.
 A) K B) L C) M D) N

CHEMICAL BONDING

1. In sp^2 hybridization bond angle is _____
 A) 120° B) 180° C) 109.5° D) None
2. $SnCl_2$ have _____ shape
 A) Planer B) Tetrahedral C) Angular D) None
3. When elements of group I-A react with the elements of group VI-A they form
 A) Ionic Bond B) Covalent Bond C) Polar Bond D) None
4. The bond between boron trifluoride and ammonia is
 A) Ionic Bond B) Covalent Bond C) Coordinate Covalent Bond D) Polar Bond
5. Which of the following has highest bond order
 A) O_2 B) O_2^+ C) O_2^- D) None
6. The sp^3 orbitals are
 A) Co-planer B) Non-Planer C) Linear D) None
7. Fused ionic compounds
 A) Insulators B) Used as Semiconductors
 C) Conduct Electricity D) Do not conduct electricity
8. The bond formed by pairing of electrons spin caused by the axial overlapping of half filled orbitals of two atoms is called
 A) Sigma Bond B) Pi-Bond C) Ionic Bond D) None
9. The bond angle depends upon the
 A) Types of Bond B) Number of Bonds
 C) Non-Bonding Electron Pairs D) All Of These
10. Which explain the overlapping of the bonding atomic orbitals
 A) VSEPR Theory B) VB Theory C) MO Theory D) All
11. The degree of polarity of molecules is known as its
 A) Dipole Moment B) Moment Arm C) Bond Energy D) Ionic Character
12. The covalent bond are
 A) Unidirectional B) Bi-Directional C) Non-Directional D) Multi-Directional
13. The covalent compounds in non-polar solvents are _____ - conductors of electricity.
 A) Good B) Bad C) Best D) Excellent
14. The bond pair electrons remain between two nuclei and are called
 A) Valance B) Stable C) Localized D) none
15. The ionic bonds are
 A) Unidirectional B) Bi-Directional C) Non-Directional D) Multi-Directional
16. Elements of group I and II combine with the elements of group VI and VII to form
 A) Ionic Bond B) Covalent Bond C) Polar Bond D) Polar Covalent Bond
17. Formation of MgO is an example of
 A) Ionic Bond B) Covalent Bond C) Polar Bond D) Double Covalent Bond
18. The properties of a substance are determined in a part by
 A) Ionic Bond B) Covalent Bond C) Hydrogen Bond D) Chemical Bond
19. A covalent bond may be
 A) 100% Covalent B) Partially Ionic C) Both A & B D) None
20. The energy of the non-bonding molecular orbital is _____ than bonding molecular orbital
 A) Lesser B) Greater C) Equal D) None
21. If two lone pairs are present then bond angle of tetrahedral compound reduces of _____ degrees.
 A) 109.5° B) 107.5° C) 104.5° D) None

22. The electro-negativity different of H-Cl is 0.9 so it should be _____ bond.
A) Covalent B) Ionic C) Coordinate Covalent D) Metallic
23. The VSEPR theory explains the _____ of molecules.
A) Number B) Kinds C) Geometry D) None
24. The Electro-negativity difference for ionic bond must be greater than
A) 1.6 B) 1.7 C) 1.8 D) 1.0
25. The polarity of molecule is expressed by
A) Bond Strength B) Dipole Moment C) Bond Length D) Shape
26. The bond between H ----- H is
A) Stronger than the bond between H ----- Cl
B) Weaker than the bond between H ----- Cl
C) Neither stronger nor weaker than the bond between H ----- Cl
D) None of thee
27. The $\text{C} \equiv \text{C}$ bond length in ethyne is
A) 154 pm B) 133 pm C) 120 pm D) 106 pm
28. The C ----- C bond length in ethane (C_2H_4) is
A) 154 pm B) 133 pm C) 120 pm D) 105 pm
29. The C ----- C bond length in ethane (C_2H_6) is
A) 154 pm B) 133 pm C) 120 pm D) 105 pm
30. The bond order of O_2 molecules is
A) 1 B) 2 C) 3 D) Zero
31. The bond order of He_2 molecules is
A) Zero B) $\frac{1}{2}$ C) 1 D) 2
32. In which of the following theories the hybridization is considered
A) VSEPR B) Lewis C) Molecule Orbital D) Valance Bond
33. Which of the following bonds is least polar?
A) H – Se B) P – Cl C) H – Cl D) N – Cl
34. Which of the following molecules has a net dipole moment?
A) CO_2 B) CS_2 C) SO_2 D) CCl_4
35. Which of the following involve ionic bonding only?
A) Li_3N B) NaCl C) NCl_3 D) O_2
36. Nitrogen molecular contains _____ bond.
A) 1σ & 1π B) 2σ & 2π C) 1σ & 2π D) 2σ & 1π
37. The ion that is isoelectronic with CO is _____.
A) CN^- B) O_2^- C) O_2^+ D) N_2^+
38. Number of σ and π bond in benzene resonance structure is _____.
A) 6σ & 6π B) 12σ & 6π C) 6σ & 3π D) 12σ & 3π
39. The octet rule is not valid for _____ molecule.
A) CO_2 B) CO C) H_2O D) CH_4
40. The theory which states that electrons pairs are arranged around the central atom od a molecule with maximum separation is _____.
A) VSEPR B) V.B.T C) M.O.T D) All
41. Dipole moment of CO_2 is zero; its structure will be:
A) Angular B) Linear C) Tetrahedral D) Octane
42. H-Cl involves _____ overlapping
A) s-s B) s-p C) p-p D) p-d
43. The bond angle in C_6H_6 is _____.
A) 109.5° B) 120° C) 180° D) 104.5°

THERMOCHEMISTRY

1. The amount of heat required to convert one mole of solid directly into its vapour state at STP is called as.
A) Molar heat of sublimation B) Standard heat of vaporization
C) Heat of Reaction D) Heat of neutralization
2. The heat of combustion is measured by
A) Calorimeter B) Colorimeter C) Bomb Calorimeter D) None
3. The heat of formation of graphite and P (white) is _____ KJ/mole
A) 0.00 B) -273.0 C) +8.7 D) 813.99
4. Spontaneous reaction is such in which the system decreases its _____
A) Energy B) Free Energy C) Entropy D) All
5. Which is used to measure heat of reaction
A) Calorimeter B) Colorimeter C) Bomb Calorimeter D) None
6. For an endothermic reaction the value of ΔE is denoted by
A) Positive Sign B) Negative Sign C) Without any Sign D) None
7. The branch of science which deals with the energy transformation is called
A) Thermochemistry B) Thermodynamics C) Chemical Kinetics D) All
8. The measurement of degree of disorder is
A) Internal Energy B) Enthalpy C) Entropy D) None
9. The amount of heat exchange when molar quantities of reaction and products are used.
A) Heat of sublimation B) Heat of neutralization
C) Heat of reaction D) None
10. Hess's law is a special case of
A) First Law of the thermodynamics B) Second Law of thermodynamics
C) Third Law of the thermodynamics D) None
11. The heat contents of all the element in their standard states are taken to be.
A) 1 B) 2 C) 0 D) None
12. The amount of heat evolved or absorbed by keeping reactants and products at one atmospheric pressure at room temperature is called
A) Heat of formation B) Standard heat of formation
C) Standard enthalpy change D) None
13. The heat content of system is called
A) Internal Energy B) Enthalpy C) Entropy D) None
14. $\Delta E = q + w$ is the
A) First Law of the thermodynamics B) Second Law of thermodynamics
C) Third Law of the thermodynamics D) None
15. Which is not state function?
A) Temperature B) Enthalpy C) Entropy D) Internal Energy
16. One joule is equal to
A) 1 Nm B) 1 Kg / s C) 1 Kgm² / s² D) Both A and C
17. Evaporation of water is _____ change
A) Endothermic B) Exothermic C) Non-energetic D) None
18. One calorie is equal
A) 4.132 J B) 760 J C) 4.184 J D) 1 J
19. An property which depends upon the T, P and V is said to be
A) Property due to K.E. B) Property due to P.E. C) Both a and b D) Thermodynamic State
20. The sum of all the energy of atoms, molecule, ion, within system is called
A) I. E. B) P. E. C) K. E. D) All
21. Any substance undergoing physical or chemical change is said to be.
A) Surrounding B) System C) Losophere D) Lithosphere
22. The energy of the reactants is not equal to the energy of the product this loss energy is in the form of
A) Heat B) Electron C) Any form of the energy D) None
23. When ammonium chloride dissolve in water this process is

- A) Endothermic Process
C) Simple Hydration
- B) Exothermic Process
D) None of the above
24. Lattice energy of NaCl
A) +5000 KJ B) -344 J C) -776 KJ D) -411 KJ
25. The common for standard enthalpy change is
A) 1 atm 30° C B) 1 atm 50° C C) 1 atm 25° C D) 760 atm 25° C
26. Pressure volume work is
A) $P \Delta v$ B) $F \times d$ C) $h \Delta v$ D) $H + pv$
27. In endothermic reaction, the heat content of the
A) Products is more than that of reactants B) Reactants is more than that of products
C) Both A and B D) None of the above
28. If an endothermic reaction is allowed to take place vary rapidly in the air. The temperature of the surrounding air
A) Remains constant B) Increase C) Decreases D) Remain Unchanged
29. Enthalpy of neutralization of all the strong acids and strong bases is the same value because
A) Neutralization lead to the formation of salt and H_2O
B) Strong acid and base are ionic substances
C) Acids always gives rise H^+ ions and bases always furnish OH^- ions
D) The net chemical change involve the combination of H^+ and OH^+ ions to form water
30. The net heat change in chemical reaction is same whether it is brought in two or more different ways in one or several steps. It is known as
A) Henry's Law B) Joule's Principle
C) Hess's Law D) Law of Conservation of energy
31. For given process, the heat changes at constant pressure (P) and constant volume (v) are related to each other as
A) $\Delta E = q + w$ B) $E = q + w$ C) $\Delta H = \Delta E + p\Delta V$ D) $\Delta H = \Delta E$
32. A gas does 234J of work at the same time it absorbs 324 J of heat the change in internal energy is
A) 98 J B) 90 J C) 558 J D) 88 J
33. Which of the following is/are intensive properties?
A) Volume B) Density C) Specific Heat D) B and C
34. Heat energy evolved in which one of the following process ($\Delta = -ve$)?
A) $2H_2 + O_2 \rightarrow$ B) $2H_2 \rightarrow 2H^+ + 2e^-$ C) $H_2O + Salt \rightarrow$ D) None of these

CHEMICAL EQUILIBRIUM

1. If the concentration of the reactants is increased then the equilibrium will
 - A) Shift towards the forwards direction
 - B) Shift towards the backwards direction
 - C) Not be affected
 - D) None
2. If $K_c > 1$ then the reaction
 - A) Moves in forwards direction
 - B) Moves in backwards direction
 - C) It is equilibrium
 - D) None
3. If we increase the pressure in decomposition of PCl_5 it will favor the reaction in
 - A) Moves in forwards direction
 - B) Moves in backwards direction
 - C) It is equilibrium
 - D) None
4. According to the Le-Chatelier's principle when heat is given to solid, which is in equilibrium with liquid
 - A) The amount of solid decreased
 - B) The amount of liquid decreases
 - C) Temperature will increase
 - D) temperature will fall
5. The Le-Chatelier's principle corresponds with the Newton's _____ Law.
 - A) First
 - B) Second
 - C) Third
 - D) Gravitation
6. In reaction $A + B \rightleftharpoons C + D$ if the concentration of A and B is double then the reaction rate will
 - A) Increase 4 times
 - B) Increase 2 times
 - C) Decrease to Half time
 - D) Will decrease 2 times
7. If a system at equilibrium is disturbed the system will try to undo the effect, this is called
 - A) Le-Chatelier's
 - B) Law of mass action
 - C) Hess's Law
 - D) Law of heat of formation
8. The direction of the reaction is predicted
 - A) Law of mass action
 - B) K_c
 - C) Le-Chatelier
 - D) Hess's Law
9. In $2\text{SO}_2 + \text{O}_2 \rightleftharpoons$ the yield of product will be maximum if
 - A) T is increased and P remains same
 - B) T is decreased and P is increased
 - C) Both T and P are increased
 - D) Both T and P are decreased
10. In the formation of ammonia if pressures of the system is increased then the reaction _____ direction.
 - A) Moves in forward
 - B) Moves in backward
 - C) Remains in equilibrium
 - D) None
11. A chemical equilibrium in reaction is established when
 - A) Concentration of the reaction is established when
 - B) Opposing reaction is happening
 - C) Velocities of the opposing reactions are equal
 - D) Temperature of the opposing reaction are equal
12. If K_c equal the concentration ratio the reaction is
 - A) Going in forward direction
 - B) going in backward direction
 - C) In equilibrium
 - D) None
13. If K_c is greater than unity then the reaction is going in the _____ direction.
 - A) Forward
 - B) Backward
 - C) Equilibrium
 - D) None
14. Rate constant for gaseous reaction is written as
 - A) K_p
 - B) K_c
 - C) K_a
 - D) K_n
15. The equilibrium established between the two different phases is called.
 - A) Homogenous equilibrium
 - B) Isogenies Equilibrium
 - C) Heterogeneous equilibrium
 - D) None
16. If K_c is very small then the product formed is in
 - A) Small amount
 - B) Large amount
 - C) Normal amount
 - D) None
17. If the temperature in the synthesis of ammonia is decreased. The formation of ammonia is
 - A) Increased
 - B) Decreased
 - C) Not effected
 - D) Side reaction takes place
18. The unit of K_c for reaction $\text{N}_2 + \text{O}_2 \rightleftharpoons 2\text{NO}$
 - A) mol dm^{-3}
 - B) mol dm^3
 - C) $\text{mol}^2 \text{dm}^{-3}$
 - D) None Units
19. Law of mass action was presented by
 - A) Guldberg & Wage
 - B) Milikon
 - C) Thomson
 - D) Ramsay
20. The sum of pH and p^{OH} is
 - A) 2
 - B) 7
 - C) 14
 - D) 11
21. Which one of the following has highest pH

- A) 0.1 M HCl B) 1.0 M HCl C) Gastric Juice D) Lemons
22. The solution having zero pH will
A) Basic B) Highly Basic C) Neutral D) Highly Acidic
23. For which reaction the numerical value of K_c and K_p are same
A) $N_2 + 3H_2 \rightleftharpoons 2NH_3$ B) $2SO_2 + O_2 \rightleftharpoons 2SO_3$
C) $H_2 + Cl_2 \rightleftharpoons 2HCl$ D) $N_2O_4 \rightleftharpoons 2NO_2$
24. Which of the following solution have zero pH
A) 1 M HCl B) MH_2SO_4 C) 0.1 M HNO_3 D) 1 M CH_3COOH
25. The precipitation occurs if the ionic concentration is
A) Less than K_{sp} B) More than K_{sp}
C) Equal K_{sp} D) Is present at any moment
26. The relation between K_c and K_p is
A) $K_c = K_p(RT)^{\Delta n}$ B) $K_p = K_c(RT)^{\Delta n}$ C) $K_c = K_p(RT)^{\Delta n}$ D) $K_p = K_c(P/N)^{\Delta n}$
27. For $N_2 + 3H_2 \rightleftharpoons 2NH_3$
A) $K_c = K_p$ B) $K_p = K_cRT$ C) $K_p = K_c(RT)^2$ D) $K_p = K_c(RT)^{-1}$
28. The solubility product of AgCl is $2.0 \times 10^{-10} \text{ mole}^2 \text{ dm}^{-6}$ the max concentration of Ag ions in the solution is
A) $2.0 \times 10^{-10} \times 10^{-10} \text{ mol dm}^{-3}$ B) $1.41 \times 10^{-5} \text{ mol dm}^{-3}$
C) $1.0 \times 10^{-10} \text{ mol dm}^{-3}$ D) $4.0 \times 10^{-28} \text{ mol dm}^{-3}$
29. The pH of $10^{-3} \text{ mol dm}^{-3}$ of an aqueous solution of H_2SO_4 is
A) 3.0 B) 2.7 C) 2.0 D) 1.5
30. For which system does the equilibrium constant, K_c has units of concentration
A) $N_2 + 3H_2 \rightleftharpoons 2NH_3$ B) $H_2 + I_2 \rightleftharpoons 2HI$
C) $CO + H_2O \rightleftharpoons CO_2 + H_2$ D) $2HF \rightleftharpoons H_2 + F_2$
31. The ionization constant of an acid is expressed in terms of the following constant
A) K_w B) K_n C) K_a D) K_b
32. Buffer action can be explained by
A) Common ion effect B) Law of mass action Le-Chatelier's principle
C) All Above D) Ionization of weak acid
33. Base buffer solution can be prepared by mixing
A) Weak acid and its salt B) Strong acid and its salt with weak base
C) Weak base and its salt with strong acid D) Strong base and its salt with weak acid
34. Units of K_w are
A) mole dm^{-3} B) $\text{mole}^2 \text{ dm}^{-3}$ C) $\text{mole}^2 \text{ dm}^{-6}$ D) $\text{mole}^2 \text{ dm}^{-3}$
35. Strength of an acid can be determined by
A) p^{K_a} B) p^{K_p} C) p^{OH} D) p^{K_w}
36. A reaction is reversible because
A) Reaction are reversible B) Products are reversible
C) Products are stable D) Reactants are stable
37. 1.0g of ethyl alcohol and 1.0g of acetic acid are mixed. At equilibrium 0.667g of ester are formed the value of equilibrium constant is _____.
A) 1 / 4 B) 1 / 2 C) 2 D) 4
38. At equilibrium
A) Conc. Of reaction and product are equal
B) Reaction rate decrease
C) Rate of forward and reverse reaction are equal
D) All are correct

SOLUTIONS

1. Which is independent of temperature?
A) Molarity B) Molality C) Normality D) Mole Fraction
2. If two or more than two substance appear to have nearly equal strengths in same solvent then the phenomenon is called.
A) Reduction B) Oxidation C) Solvation D) Leveling Effect
3. Solution which contains the hydrogen ions more than the 10^{-7} M is said to be.
A) Acidic B) Basic C) Crystal D) None
4. The substance which contains the water of crystallization is called.
A) Hydrate B) Solvated C) Crystal D) None
5. The molarity of solution made by the 98g of sulfuric acid in sufficient water to make 500 ml of solution is
A) 1 M B) 0.5 M C) 2 M D) 3 M
6. The process in which the solvent molecules are surrounded and interact with solute ions or molecules is called.
A) Solvation B) Hydration C) Hydrogenation D) None of these
7. The ratio of number moles of a particular component of the solution to the total number of moles of all the components is called.
A) Normality B) Molarity C) Molarity D) Mole Fraction
8. The amount of solute present in the given amount of solvent is called.
A) Molarity B) Molality C) Concentration D) Solubility
9. A solution can be
A) Dilute and Concentrated B) Saturated and Dilute
C) Saturated and Unsaturated D) Supersaturated and Saturated
10. Number of moles of solute per Kg of solvent is called:
A) Molarity B) Molality C) Mole Fraction D) None
11. Number of moles of solute per liter of solvent is called:
A) Molarity B) Molality C) Mole Fraction D) None
12. Solution may have units.
A) Molarity B) Molality C) Mole Fraction D) All of them
13. Homogenous mixture of two chemical substance, whose proportion can be varied is called.
A) Solution B) Mixture C) Binary Solution D) Saturated Solution
14. Colligative properties are the properties of solution that depend upon
A) Nature of Molecules B) Unity
C) Physical Property D) None of above
15. The no of water of crystallization of $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$
A) 12 B) 6 C) 3 D) 4
16. Which one of the following is not soluble in alcohol
A) KCl B) Urea C) Acetone D) Ether
17. When NaCl is dissolved in water
A) Melting point Decreases B) Boiling
C) Both melting and boiling points, decrease D) None of above
18. The sum of mole fractions (X) of components of solution is equal to
A) 100 B) 200 C) One D) Zero
19. Which one of the following salts will not hydrolyze.
A) NaCl B) AlCl_3 C) Na_2CO_3 D) CH_3HOONa
20. Hydrolysis of potassium acetate produces.
A) Acidic Solution B) Neutral Solution C) Basic Solution D) None of these
21. A solution of sucrose is 34.2% the volume of solution containing one mole of solute
A) 500 cm^3 B) 1000 cm^3 C) 342 cm^3 D) 3420 cm^3
22. A solution of glucose is 10%. The volume in which in 1 gram mole of it is dissolved will be
A) 1 dm^3 B) 1.8 dm^3 C) 200 dm^3 D) 900 dm^3
23. Which of the following solution has the highest elevation in boiling point.
A) 5.85% solution of NaCl B) 18.0% solution of glucose
C) 6.0% solution of Urea D) All have same boiling of glucose
24. 4.0 NaOH has been dissolved per dm^3 of the solution. The strength of the solution is _____.
A) 0.5 M B) 0.6 M C) 0.1 M D) 1.0 M
25. What is the required volume of 0.1 M NaOH to neutralize 2 ml of 0.5 M H_2SO_4 ?

- A) 10 ml B) 20 ml C) 2 ml D) 50 ml
 26. The percentage of HCL in gastric juice is _____.
 A) 1.5% B) 2% C) 0.5% D) 3%
 27. The acidic buffer solution is _____.
 A) $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$ B) $\text{CH}_3\text{COOH} + \text{CH}_3\text{COONa}$
 C) $\text{NH}_4\text{OH} + \text{CH}_3\text{COONH}_4$ D) $\text{NaOH} + \text{HCl}$

ELECTROCHEMISTRY

1. In electrolytic cell oxidation occurs at
 A) Anode B) Cathode C) During Movement D) None
2. Zinc copper galvanic cell may be formed by
 A) Salt Bridge B) Porous Partition C) Both A and B D) None
3. The oxidation number of free element is always taken
 A) 0 B) 1 C) 2 D) -1
4. The tendency of an element to form ions is always taken.
 A) Cell Potential B) Ionization Potential
 C) Electrode Potential D) None
5. A cell in which the electrical current is passed through the redox reaction is called
 A) Dry Cell B) Electrolytic Cell C) Voltaic Cell D) None of these
6. In SHE the standard is the _____ atom.
 A) H B) C C) N D) O
7. The value of SHE is cathode and anode is always taken to be
 A) One B) Zero C) Different D) Same
8. Strongest reducing agent in the electrochemical series is
 A) Li B) F C) H_2 D) I_2
9. Corrosion reactions are
 A) Spontaneous redox reaction B) Non-spontaneous acid-base reactions
 C) Spontaneous acid-base reaction D) None of these
10. Metals which are above SHE in electrochemical series
 A) Can liberate H_2 from acid B) Cannot liberate H_2 from acid
 C) Cannot always liberate H_2 from acid D) None of these
11. Strong oxidizing agent have
 A) Greater positive value of standard reduction potential
 B) Lesser Positive value of standard reduction potential
 C) Greater negative value of standard reduction potential
 D) None of these
12. Strong reducing agents have
 A) Greater positive value of standard reduction potential
 B) Lesser Positive value of standard reduction potential
 C) Greater negative value of standard reduction potential
 D) None of these
13. In alkaline battery, the electrolyte contains
 A) MnO_2 B) KOH C) NaCl D) NaNO_3
14. In lead accumulator the electrolyte is H_2SO_4 solute is
 A) 30% H_2SO_4 B) 60% H_2SO_4 C) 80% H_2SO_4 D) 90% H_2SO_4
15. Greater the value of standard reduction potential of species indicates
 A) Greater its tendency to accept electrons
 B) Lesser tendency to accept electrons
 C) Greater tendency to lose electrons
 D) None of these
16. A process for converting one metal with thin layer of another metal is called
 A) Electrolysis B) Electroplating C) Electrode Potential D) Standard
17. Cell in which electric current drives a non-spontaneous reaction is called
 A) Electrolytic Cell B) Voltaic Cell C) Biological Cell D) Electrochemical Cells
18. A cell in which spontaneous redox reaction generates an electric current is called

- A) Electrolytic Cell B) Electrochemical Cell C) Voltaic & Galvanic Cell D) Biological Cell
19. The branch of chemistry which deals with the relationship between electricity and chemical reaction is called
A) Chemical Kinetics B) Electrochemistry C) Stiochiometry D) Hermochemistry
 20. Sodium metal is obtained by the electrolysis defused NaCl in cell is called
A) Nelson's Cell B) Down's Cell C) Daniel Cell D) Voltaic Cell
 21. Caustic soda is obtained by electrolysis of concentrated aqueous solution of NaCl in a cell called
A) Daniel's Cell B) Nelson's Cell C) Down's Cell D) Voltaic Cell
 22. The reaction in galvanic cell is
A) Spontaneous B) Non-spontaneous C) Acid-base D) None of these
 23. The metallic conductors in contact with the solution are called
A) Insulator B) Electrodes C) Electrolyte D) Down Cell
 24. The process in which ionic compound when fused or dissolved water splits up into charged
A) Electrolyte B) Hydration C) Ionization D) Conduction
 25. The process in which electric current is used to carry out a non-spontaneous redox reaction is called
A) Electrolyte B) Electrolysis C) Metallic Conductor D) None of these
 26. Oxidation number of oxygen in OF_2 is
A) +6 B) -2 C) +2 D) +4
 27. Electric current passed through both molten and solution form of NaCl because of
A) Ionic Bonding B) Na^+ and Cl^- ions C) Ions of water D) Hydration of Ions
 28. Which element acts as a reducing agent in the reaction $\text{Zn} + \text{H}_2\text{SO}_4 \rightleftharpoons \text{ZnSO}_4 + \text{H}_2$
A) Zn B) H C) S D) O
 29. During electrolysis of H_2SO_4 the O_2 is evolved at
A) Cathode B) Anode C) Both A and B D) None of these
 30. Oxidation number of oxygen in OF_2 is
A) +1 B) -1 C) +2 D) -2
 31. In a oxidation process the oxidation number of the element
A) Increases B) Decreases C) Dose not change D) None of these
 32. During redox reaction an oxidizing agent
A) Gain electrons B) Is Oxidized C) Loses Electrons D) Hydrolyzed
 33. Which of the following will be good conductor of electricity
A) Pure distilled Water B) Molten NaCl
C) Dilute Solution Of Glucose D) Chloroform
 34. An electrolysis of KNO_3 , H_2 is evolved
A) Acid-base reaction B) Redox Reaction C) Nuclear Reaction D) None of the above
 35. During electrolysis of KNO_3 , H_2 is evolved
A) Anode B) Cathode C) Both A and B D) None
 36. When fused PbBr_2 is electrolyzed then
A) Bromine appears at cathode B) Lead deposited at the cathode
C) Lead appears at the anode D) None of these happens
 37. In the reaction $2\text{Fe} + \text{Cl}_2 \rightleftharpoons 2\text{FeCl}_3$
A) Fe is reduced B) Fe is oxidize C) Cl_2 is oxidized D) None of these
 38. The Oxidation number of chromium in $\text{K}_2\text{Cr}_2\text{O}_7$ is.
A) 14 B) 12 C) 12 D) None of these

REACTION KINETICS

1. $3\text{Fe} + 4\text{H}_4 \rightleftharpoons \text{Fe}_3\text{O}_4 + 4\text{H}_2$ is an example of _____ equilibrium.
A) Homogeneous B) Heterogeneous C) Isogeneous D) None
2. _____ are called biocatalysts.
A) Organic Acids B) Organic bases C) Enzymes D) All
3. The minimum function of a catalyst is to
A) Increase E_a B) Decrease E_a C) Decrease Temp D) Decrease Pressure
4. The rate of a reaction depends upon
A) Surface Area B) Temperature C) Concentration D) All
5. The reaction may be
A) Third Order B) Four Order C) Both A and B D) None
6. The reaction rate can be measure by
A) Chemical Method B) Conduct Metric Method C) Both A and B D) None
7. The units of rate constant of first order reaction are
A) S B) S^{-1} C) Mole S D) None
8. The addition of a catalyst use to decrease
A) Enthalpy B) Entropy
C) Nature of Reactants D) Energy of activation
9. The energy of activation is usually expressed in
A) Ergs B) Calories C) Mole D) Joules
10. The actual number of atoms or molecules taking part in rate determining step is
A) Rate of reaction B) velocity of reaction C) Order of reaction D) None
11. The chemical reactions which may proceed slowly with measurable rate are called
A) slow reactions B) fast reaction C) moderate Reaction D) None
12. An increase in conc. is related to number of collisions
A) Directly B) indirectly C) No more D) All
13. The reaction rate is expressed in the units of
A) $\text{mol dm}^3 \text{ s}^{-1}$ B) mold m^{-3} C) $\text{mold m}^{-3} \text{ N}^{-1}$ D) $\text{dm}^{-3}\text{s}^{-1}$
14. The change in the conc. of reactants and products in unit time is called
A) Reaction rate B) Rate Constant C) Diffusion D) None
15. For forward reaction according to collision theory the molecules must have energy.
A) Greater than E_a B) Less than E_a C) Equal to E_a D) None
16. The rate of reaction may be
A) Zero Order B) Never Zero Order C) Without any Order D) None
17. The number of atoms molecules ions whose conc. determinate the rate reaction is called
A) Rate Constant B) Value or reaction C) Order of reaction D) None
18. Decrease in concentration of reaction is denoted by
A) Dc/dt B) $-\text{dc/dt}$ C) $+\text{dc/dT}$ D) None
19. The rate of reaction is denoted by
A) Dc/dp B) dc/ac C) dc/dT D) dx/dt
20. The branch of chemistry which deals with the study of rate of reaction is called
A) Thermodynamic B) Chemistry of Reversibility
C) Chemical Kinetics D) Mechanistic Studies
21. Catalysis may be
A) Homogeneous B) Heterogeneous C) Both A and B D) None
22. By the use of catalysis the energy of activation is
A) Decreased B) increased C) Not Affected D) None
23. The reaction rate may be measured by
A) Physical Method B) Chemical Method C) Both A and B D) None
24. The factors which effect the rate of reaction
A) Nature of Reactants B) Surface Area C) Light D) All of these
25. In the hydrolysis of $\text{CH}_3\text{COOC}_2\text{H}_5$ the acid produced is
A) Inhibitor B) Catalyst C) Auto Catalyst D) None of these
26. Optical rotation method is used when
A) Reaction Involve Ions B) Change of Refractive Indices
C) Reaction involving change of optical activity D) None of the above
27. Refractometric method is used when
A) Reactions involving absorption I.R. or U.V. B) Reactions involving change of refractive index

- C) Reaction Involve Ions D) Change of optical activity
28. Electrical conductivity method is applied for rate determination when
 A) Reactions and products involve absorption of U.V. or I.R radiation
 B) Reaction Involve Ions
 C) Reaction which involve Change in refractive indices
 D) Reaction which involve small volume change
29. Spectrophotometry is applied for rate determination when
 A) Reactions or products, absorb U.V. and I.R. light
 B) Reaction Involve Ions
 C) Reaction involve Change in volume
 D) None of above
30. When reaction occurs in many steps then the slowest step is the
 A) Min Step B) Enthalpy Determining step
 C) Mechanism determining step D) Rate determining step
31. When the rate of reaction is entirely independent of the conc. of reactant molecules then order of reaction is
 A) Zero B) First C) Second D) Third
32. Photochemical reaction usually have order
 A) One B) Zero C) Two D) Three
33. The sum of the exponents of the conc. terms in the rate equation is called
 A) Rate of Reaction B) Order of Reaction C) Specific Rate Constant D) Average rate
34. The rate of reaction between two specific time intervals is called
 A) Instantaneous Rate B) Average Rate C) Specific Rate D) Ordinary Rate
35. With increase of 10°C temperature the rate of reaction doubles. This increase in the rate of reaction is due to
 A) Decreasing in activation energy of reaction
 B) Decreasing in the number of collision between reactants molecules
 C) Increase in activation energy of reactants
 D) Increase in number of effective collisions.
36. The rate of reaction
 A) Increase as the reaction proceeds
 B) Decreasing as the reaction proceeds
 C) Remains the same as the reaction proceeds
 D) May decrease or increase as the reaction proceeds.
37. A + B Product

A	B	Rate
1.3	2.2	0.04
1.3	3.3	0.06
2.6	3.3	0.24
2.6	2.2	0.16

Which of the following statement is false?

- A) 2nd order in B B) 2nd Order in A
 C) Rate = K [A]² [B]¹ D) 3rd Order reaction
38. The energy of activation in the absence of catalyst is 75 KJ/mol. The most likely value of energy of activation in the presence of catalyst will be _____.
 A) 53 KJ/mol B) 75 KJ/mol C) 105 KJ/mol D) 175 KJ/mol

PERIODIC CLASSIFICATIONS OF ELEMENTS AND PERIODICITY

1. The %age of C in different types of soft iron products is in the order of
 A) Oxidation B) Reduction C) Hydrogenation D) Dehydrogenation
2. From ³⁹Y to ⁴⁸Cd are called
 A) Transition Element B) Outer Transition Elements
 C) Inner Transition Elements D) 2nd Transition Series
3. The electropositive elements from
 A) Acidic Oxides B) Basic Oxides C) Neutral Oxides D) None
4. The units of ionization energy are
 A) J B) Cal C) E.V. D) None

5. Which is the most volatile compound
A) HI B) HCl C) HBr D) HF
6. Elements in the same family have
A) Same atomic number B) Molecular wt same
C) Same chemical properties D) Same electronic Configuration
7. The lowest ionization energy are found in
A) Inert gases B) Alkali metals C) Alkali earth metals D) Halogens
8. Which of the following is a transition element
A) Zr B) Ru C) Rh D) All
9. In periods the elements are arranged strictly according to their
A) Increasing atomic wt B) Increasing Nuclear Charge
C) Stable Elements D) Increasing Valency
10. All the elements belonging to the 2nd period are
A) Typical Element B) Transition Element
C) Stable Element D) Halogens
11. A pair of elements in the same family in the periodic classification is
A) Cl and C B) Ca and Al C) N and Ne D) Na and K
12. Excluding He and H the smallest element in periodic table is
A) Li B) F C) Cs D) I
13. The coinage metals are
A) Ni, Pd, Pt B) Cu, Ag, Au C) Zn, Al, Pb D) Fe, Si, Sn
14. In a group from top to bottom, the hardness of alkali metals
A) Remains Unchanged B) Increases
C) Decreases D) None
15. The correct order of 2nd I.P. of C, N, O and F is
A) $O > F > N > C$ B) $O > N > F > C$ C) $C > N > O > F$ D) $F > O > N > C$
16. Na_2O is
A) Acidic B) Basic C) Neutral D) Amphoteric
17. Ionization energy depends upon
A) Nuclear B) Atomic Size C) Shielding D) All of above
18. In a period the alkali metals have
A) Smallest atomic Size B) Lowest I.E.
C) Lowest Density D) Highest E.A.
19. Which has smallest ionic radius
A) K^+ B) Ca^{2+} C) Sc^{+3} D) Ti^{4+}
20. Which element has highest density
A) Li B) Be C) C D) Ne
21. In sixth period 14 of its transition elements are called
A) Lanthanides B) Actinides C) Radioactive Elements D) None
22. Sixth period contains maximum _____ electrons in its valence shell.
A) 10 B) 8 C) 18 D) 32
23. Element of 1st period are also called
A) S-block B) P-block element C) D-block elements D) F-block element
24. Ga belong to _____ family.
A) Boron B) Carbon C) Nitrogen D) Beryllium
25. Periodic law was given by
A) Al-Razi B) Dobriener C) Newland D) Mendeleev
26. Debiere arranged elements in a group of
A) Duplets B) Triads C) Pentaves D) Hexaves
27. There was seven groups in periodic table proposed by:
A) Al-Razi B) Dobriener C) Newland D) Mendeleev
28. Mendeleev periodic table was arranged in the order of atomic masses:
A) Decreasing B) Increasing C) Decreasing D) Descending

29. An extra group of noble gases (VIII-A) was added in table of:
 A) Al-Razi B) Mendeleev C) Newland D) Mosley
30. Vertical column in periodic table are called:
 A) Groups B) Periods C) Blocks D) Lanthanides
31. Modern periodic table has
 A) Eight Group and Seven Periods B) Eight Group and Seven Periods
 C) Eight Group and Eight Periods D) Seven Group and Seven Periods
32. Short periodic contain elements:
 A) 2 B) 8 C) 18 D) 32
33. A radius smaller than parent atom is called:
 A) Atomic B) Covalent Radii C) Ionic Radii D) Cationic Radii
34. Size of an atoms is measured:
 A) Directly B) Indirectly
 C) May Be Directly D) May Be Directly or Indirectly
35. Atomic radius decreases in periods because of:
 A) Increases in nuclear charge B) No effect of Shielding
 C) Increases in nuclear group D) a, b, c All are factors
36. The size of atom in positive ion decreases because:
 A) Outermost Shell Loses B) Imbalance no. Protons
 C) A & B D) Shielding Effect Increases
37. The minimum amount of energy required to dissociate a crystal lattice into its is called:
 A) Ionization Energy B) Electron Affinity C) Electron Negativity D) Lattice Energy
38. Trend of ionization energy in a period from left to right is:
 A) Increases B) Decreases C) Remains Same D) Increases
 Decreases
39. Metals are present generally in periodic table:
 A) Left Hand Side B) In the center and bottom
 C) A & B D) Right Hand Side
40. Properties of metals is/are.
 A) Loose Electrons B) Good Conductor
 C) Basic Oxides and Hydroxides D) All Above

S-BLOCK ELEMENTS

1. Li Forms
A) Organo Metallic Compound B) Coordination Compound
C) Both A and B D) None
2. The heat of hydration _____ from top to bottom
A) Increases B) Decreases C) Not Affected D) Remains Same
3. The formula of pairs is
A) $(\text{CaSO}_4) \cdot \frac{1}{2}\text{H}_2\text{O}$ B) $(\text{CaSO}_4) \cdot 5\text{H}_2\text{O}$ C) $(\text{CaSO}_4) \cdot 7\text{H}_2\text{O}$ D) $(\text{CaSO}_4) \cdot 9\text{H}_2\text{O}$
4. At _____ °C gypsum losses its water molecules
A) 75 B) 100 C) 125 D) 150
5. Calcium Sulphate occurs in nature in Di hydrated form called
A) Soda Ash B) Lime Soda C) Baking Soda D) Gypsum
6. Due to the explosive nature the Na metal is stored in
A) Alcohol B) Hydrogen per oxide
C) Sodium Hydroxides D) Kerosene Oil
7. The alkaline earth metals except _____ are white in color.
A) Soluble B) Insoluble C) Partially Soluble D) None
8. Alkaline earth metals form _____ carbides when heated with carbon
A) Soluble B) Insoluble C) Unstable D) Stable
9. All alkaline earth metals except _____ are white in color
A) Be B) Mg C) Ca D) None
10. The chemical formula to Trona is
A) $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ B) Na_2CO_3 C) $\text{Na}_2\text{CO}_3 \cdot \text{NaHCO}_3 \cdot \text{H}_2\text{O}$ D) KCl
11. When NaCl is dissolved in water the sodium ions are _____ during electrolysis
A) Oxidized B) Reduced C) Hydrolyzed D) Hydrated
12. Which of the following elements has highest I.P. values?
A) K B) Rb C) Cs D) Li
13. The basic strength of solution formed as result of chemical reaction of I-A and II-A is:
A) Increased down the group B) Decreasing down the group
C) Not affected D) None
14. Some of the elements of a period show similar behavior with the elements of next group in next period this is called
A) Vertical Relationship B) Oblique Relationship
C) Diagonal Relationship D) None
15. In alkaline earth metals, from top to bottom generally the ionic size is
A) Increases B) Decreases C) Not Affected D) None
16. In S-block elements from top to bottom generally the ionic size is
A) Increases B) Decreases C) Not Affected D) None
17. Na may act as
A) Oxidizing Agent B) Reducing Agent C) Both A and B D) None
18. The Chemical formula Sylvite is
A) KCl B) $\text{Na}_2\text{CO}_3 \cdot \text{H}_2\text{O}$ C) $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ D) None
19. Alkali metals act as
A) Reducing Agent B) Oxidizing Agent C) Bleaching Agent D) None
20. The formula of washing soda is
A) $\text{Na}_2\text{CO}_3 \cdot 7\text{H}_2\text{O}$ B) $\text{Na}_2\text{CO}_3 \cdot 8\text{H}_2\text{O}$ C) $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ D) Na_2CO_3
21. Which element has highest oxidation potential
A) Li B) Be C) Ba D) Ra
22. CaC_2 on hydrolysis from
A) CH_4 B) C_2H_4 C) C_2H_2 D) C_6H_6
23. Which is not mineral of alkali metals?
A) NaOH B) KOH C) LiOH D) None

24. _____ is called milk of magnesia
 A) NaOH B) KOH C) LiOH D) None
25. Which is not hygroscopic
 A) NaOH B) KOH C) LiOH D) None
26. Which oxide is most soluble in water?
 A) BeO B) MgO C) CaO D) BaO
27. Which metal is least reactive
 A) Be B) Na C) Mg D) All
28. _____ gives peroxide
 A) Li B) Ba C) Sr D) Be
29. _____ reacts with alkalis to give hydrogen
 A) Be B) Mg C) Ca D) None of the above
30. Which of the following can only form normal oxide?
 A) K B) Li C) Na D) None
31. Which of the following is the hardest metal among following
 A) Li B) Na C) Rb D) K
32. LiOH _____ soluble than NaOH.
 A) More Soluble B) Less Soluble C) Equality Soluble D) None
33. Which of the following is lighter
 A) Li B) K C) Na D) None
34. Rubidium is found in the deposit of
 A) Na B) K C) Be D) Mg
35. Electron affinity value is expressed in figures
 A) Positive B) Negative C) Sometime Positive D) Sometime Negative
36. Ionization Energy value is expressed in figures
 A) Positive B) Negative C) Sometime Positive D) Sometime Negative
37. Which one has zero oxidation state?
 A) Alkali Metals B) Alkali Earth Metals
 C) Noble Gases D) Lanthanides
38. Hydration energy of ions depend upon:
 A) Mass of Ion B) Charge to size ratio of ions
 C) P.E. of ions D) Charge of ion
39. Bleaching powder may be produced by passing chlorine over
 A) Calcium carbonate B) Hydrated Calcium Sulphate
 C) Anhydrous Calcium Sulphate D) Calcium Hydroxide
40. Which is used to remove air bubbles from metals?
 A) B B) Be C) Mg D) Al
41. Which ion will have the maximum value of heat of hydration?
 A) Na^+ B) Cs^+ C) Ba^+ D) Mg^+
42. Which of the following Sulphate is not soluble in water?
 A) Sodium Sulphate B) Potassium Sulphate
 C) Zinc Sulphate D) Barium Sulphate
43. Which oxide is insoluble?
 A) Li_2O B) Na_2O C) K_2O D) BeO
44. Dawn's cell is used to prepare:
 A) Sodium B) Sodium Hydroxide C) Calcium D) magnesium
45. Milk of lime is
 A) MgO B) CaO C) $\text{Ca}(\text{OH})_2$ D) $\text{Mg}(\text{OH})_2$
46. Which one is alkaline earth metal?
 A) Sodium B) Beryllium C) Potassium D) Lithium
47. The strong alkaline solution in water is of metal;
 A) Sodium B) Magnesium C) Aluminum D) Silicon
48. The alkaline earth metals are represented as:
 A) M^{+1} B) M^{+2} C) M^{+3} D) M^{+4}

49. Sodium mineral is
 A) Spodumene B) Halite C) Carnallite & Sylvie D) Beryl
50. Chile saltpeter is mineral of
 A) Lithium B) Sodium C) Potassium D) Beryllium
51. The amphoteric oxide is:
 A) Na_2O B) K_2O C) BeO D) MgO
52. Formula of sodium beryl late is:
 A) Na_2BeO_2 B) NaBeO_2 C) NaBe_2O D) NaBeO
53. Which is used for treatment of acidity of stomach?
 A) CaO B) BaO C) MgO D) Mg(OH)_2
54. Calcium carbonate decomposes releasing gas:
 A) H_2O B) CO_2 C) CO D) O_2
55. Hydrogen releases is prepared by the cell:
 A) Cathode B) Anode C) In Cell Basin D) Both at anodes and cathodes
56. Major problem in Nelson cell are:
 A) Down's B) Nelson C) Nelson & Diaphragm D) Diaphragm
57. Major problems in Nelson cell are:
 A) Formation of Hypochlorite B) Contamination of Cl_2 due to O_2
 C) Mixing of NaCl on NaOH D) A, B and C
58. A cell having redox process it is called:
 A) Electrolytic Cell B) Galvanic Cell C) Voltaic Cell D) Electrochemical Cell
59. Plaster of Paris is used in:
 A) Making Plaster Walls B) Status C) Coins D) All
60. Milk of lime is calcium hydroxide in water as:
 A) Mixture B) Solution C) True Solution D) Suspension
61. Which one of the following does not belong to alkaline
 A) Be B) Ra C) Ba D) Rn

GROUP III-A AND GROUP IV-A ELEMENTS

1. The hydrolysis of borax is prevented in the presence of
 A) Ethanol B) Vinegar C) Glucose D) Glycerin
2. _____ is used in making cosmetics
 A) Borax B) Koalin C) Soapstone D) Asbestos
3. Reducing flame of Ni by borax bead test is
 A) Reddish brown B) Grey C) Opaque D) Both b and c
4. Which acid is titrated in the presence of glycerin?
 A) Carbonic acid B) Boric acid C) Tartaric acid D) None of these
5. Which of the following is not a good pigment
 A) $\text{Pb(NO}_3)_2$ B) Pb_3O_4 C) $2\text{PbCO}_3 \cdot \text{Pb(OH)}_2$ D) None
6. Which of the following is a hydrated compound
 A) $\text{Mg}_3\text{H}_2(\text{SiO}_3)_4$ B) $\text{CaMg}_3(\text{SiO}_3)_4$ C) $\text{Al}_2\text{O}_3(\text{SiO}_3)_4$ D) None
7. Massicot is _____ in color
 A) Pale yellow B) Readdish brown C) Reddish Yellow D) Both a and c
8. PbO is commonly known as
 A) Red lead B) White lead C) Litharge D) None
9. Which is the ore of Al?
 A) Duralumin B) Magnetism C) Alnico D) Bauxite
10. Tincal is a mineral of
 A) Al B) B C) Si D) C
11. The chief ore of aluminium is
 A) Na_3AlF_6 B) $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$ C) Al_2O_3 D) $\text{Al}_2\text{O}_3 \cdot \text{H}_2\text{O}$
12. Sodium silicate is the sodium salt of
 A) Silica B) Quartz C) Silicic acid D) Metasilicic acid
13. Quarts is generally
 A) CaCO_3 B) $\text{MgCO}_3 \cdot \text{CaCO}_3$ C) SiO_2 D) None
14. Vitreous silica reacts with

15. A) HCl B) HBr C) HF D) All of the above
PbO behaves as a _____ oxide
16. A) Acidic B) Basic C) Neutral D) Amphoteric
The chemical formula of magnesite is
17. A) CaCO_3 B) $\text{MgCO}_3 \cdot \text{CaCO}_3$ C) MgCO_3 D) None
_____ shows inert pair earth crust is
18. A) C B) N C) O D) Si
The most abundant element of earth crust is
19. A) Na B) B C) C D) Sn
_____ forms Amphoteric oxide
20. A) Pyroboric acid B) Metboric acid C) Boric acid D) All
When we heat orthoboric acid at 100°C it is converged into
21. A) H_3BO_3 B) H_2BO_4 C) $\text{H}_6\text{B}_4\text{O}_9$ D) HBO_4
The chemical formula of pyroboric acid is
22. A) H_3BO_3 B) HBO_2 C) H_2BO_4 D) None
The formula of metaboric acid is
23. A) C B) Ge C) Sn D) Si
B has diagonal relationship with
24. A) Isomers B) Isotopes C) Allotropes D) All
Elements with two or more physical and molecular forms are called
25. A) C and Na B) C and Si C) Sn and Pb D) All of the above
Which pair forms acidic oxide
26. A) Monobasic acid B) Triacidic base C) Tribasic acid D) None of these
 H_3BO_3 is _____
27. A) HCl B) $\text{Cu}(\text{BO}_2)_2$ C) NaOH D) $\text{Na}_2\text{B}_4\text{O}_7$
_____ is used softening of water
28. A) NaBO_2 B) $\text{Na}_2\text{B}_4\text{O}_7$ C) H_3BO_3 D) $\text{Na}_2\text{B}_4\text{O}_7 \cdot 5\text{H}_2\text{O}$
Which of the following compounds has triclinic crystal system?
29. A) Cubic crystals B) Hexagonal crystals
C) Octahedral crystals D) Decahedral crystals
At 62°C borax forms
30. A) Al B) B C) Ga D) All
Which of the following cannot form stable cation?
31. A) Na_3AlF_6 B) $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$ C) NaAlO_2 D) Al_2O_3
The chemical formula of Cryolite is
32. A) 3 B) 5 C) 7 D) 9
The total no of elements in group IV-A is
33. A) C B) B C) F D) N
Which has minimum reactivity
34. A) F B) Cl C) O D) N
Boron always occurs in nature in combination with
35. A) Small size B) High nuclear charge
C) Both a and b D) None
Which factor makes boron non-metallic?
36. A) Diamond B) Graphite C) Charcoal D) Coke
Allotropic form of carbon having giant tetrahedral structure
37. A) Conductor B) Non-conductor
C) Semi-conductor D) May be conductor
 AlF_3 is a
38. A) Ionic B) Covalent C) Polymeric D) Polar covalent
 PbCl_2 is mainly
39. A) $\text{Ca}_2\text{B}_6\text{O}_{11} \cdot 5\text{H}_2\text{O}$ B) $\text{CaB}_4\text{O}_7 \cdot 4\text{H}_2\text{O}$
C) $\text{Na}_2\text{B}_4\text{O}_7 \cdot 4\text{H}_2\text{O}$ D) $\text{CaNaB}_5\text{O}_9 \cdot 8\text{H}_2\text{O}$
Chemical composition of colemanite is
40. A) A antiseptic in medicine B) For Washing eyes
C) In soda bottles D) For enamels and glazes
Boric acid cannot be used

41. Pair of element belonging to group III-A?
A) Al & Ca B) B & Be C) Ga & Ge D) In & Tl
42. Borax is used
A) Borate glasses B) In water softening C) As flux D) All of these
43. Which is used as antiseptic medicine?
A) Boric acid B) Borax
C) Sodium meta borate D) Metaboric acid
44. Pure clay is also called
A) Asbestos B) Sodium Silicate C) Kaolin D) Feldspar
45. Semi conductor is
A) Ge B) Se C) Si D) All
46. Lead dioxide paint has colour
A) White B) Black C) Reddish brown D) Yellow

GROUP V_A AND VI_A ELEMENTS

1. SO₃ is not absorbed in water directly to form H₂SO₄ because
A) Reaction does not go to completion B) Reaction is quite slow
C) Reaction is exothermic D) SO₃ is insoluble in water
2. Chemical formula of heavy spar _____
A) CaSO₄ B) BaSO₄ C) MgSO₄ D) Ca(NO₃)₂
3. _____ is called oil of vitriol
A) H₂SO₄ B) HNSO₄ C) HCl D) CH₃COOH
4. Noble metal are dissolved in _____
A) Conc. H₂SO₄ B) Aqua Regia C) Conc. HCl D) All of these
5. _____ is called nitrosyl chloride
A) NO₂Cl₂ B) NOCl C) NO(OCI) D) None
6. NO is produced by reaction of Cu with
A) Dil. HNO₃ B) Conc. HNO₃ C) Conc. H₂SO₄ D) All of these
7. P exists in _____ solid allotropic forms
A) 2 B) 3 C) 5 D) 6
8. The catenation is ability of carbon to grow in
A) Size B) Number C) Line D) Long chains
9. Which of the following does not react with HNO₃
A) Au, Pt B) Mg, Mn C) Cu, Hg D) All of these
10. Which is not true about nitric acid?
A) Have pungent smell B) Is a volatile liquid
C) Read in color D) All of these
11. Deamination is the removal of _____ group
A) NO₂ B) N₂O₄ C) NH₂ D) All of these
12. Which is true for nitrogen?
A) It is non-metal B) It has more tendency to attract electrons
C) It has 2 isotopes D) All of these
13. NO reacts with O₂ to give
A) N₂O B) N₂O₃ C) N₂O₄ D) NO₂
14. Which of the following specie has the maximum no of unpaired electrons?
A) O₂ B) O₂⁺ C) O₂⁻² D) O₂⁺²
15. Which gas greater tendency to attract electrons?
A) N B) P C) As D) Bi
16. The chemical formula of laughing gas is
A) N₂O B) NO₂ C) N₂O₅ D) None
17. Conc, sulphuric acid is used as
A) Reducing agent B) Oxidizing C) Dehydrating D) None
18. The freezing point of sulphuric acid is _____ °C
A) 10.24 B) 10.40 C) 10 D) None
19. H₂S act as a _____ agent
A) Oxidizing B) Reducing C) Neutralizing D) None
20. The most abundant element in bonded state in the earth crust is
A) C B) N C) O D) Si
21. The first four elements of group V – A are called

22. When sulphuric acid is treated with sugar it gives
 A) Halogen B) Coinage metals C) Chalcogens D) None
 A) CO_2 B) CO C) SO_2 D) None
23. The B.P. of the nitric acid is _____ $^{\circ}\text{C}$
 A) 58 B) 83 C) 71 D) 92
24. The members of V---A group show the maximum oxidation state
 A) +2 B) +4 C) +5 D) +6
25. Pyrosulphate is formed in
 A) Contact Process B) Lead Chamber Process
 C) Both A and B D) None
26. Aqua Regia is formed by mixing one volume of nitric acid and three volumes of
 A) H_2SO_4 B) HCl C) NaOH D) Any of these
27. If a metal M^{+2} reacts with oxygen then the formula of compound formed is
 A) MO B) M_2O C) MO_2 D) M_2O_2
28. PCl_5 is
 A) Conductor B) Non-Conductor C) Sublime-Solid D) May Be Conductor
29. Which pair of elements from long chains?
 A) C and Sn B) C and Si C) Si and Sn D) Sn and Pb
30. About 25% of earth crust mass is made up of element
 A) Carbon B) Silicon C) Germanium D) Tin

THE HALOGENS AND THE NOBEL GASES

1. Halogens do not react directly with
 A) Carbon B) Boron C) Oxygen D) Both a and b
2. Which one is the strongest acid
 A) HClO B) HClO_2 C) HClO_3 D) HClO_4
3. Formula of halite is
 A) KCl B) LiCl C) CsCl D) NaCl
4. _____ is used for prediction of earthquake.
 A) Helium B) Radon C) Fluorine D) Krypton
5. Solubility of noble gases _____ down the group
 A) Increases B) Remains same C) Decreases D) None
6. Which has high boiling point
 A) He B) Ne C) Ar D) Kr
7. _____ is used to prepare un-shrinkable wool
 A) CaOCl_2 B) HF C) HCl D) $\text{Ca}(\text{OH})_2$
8. In Hasenlever method chlorine reacts with
 A) OCl^{-1} B) Ca C) $\text{Ca}(\text{OH})_2$ D) None
9. In O_3F_2 the oxidation no of O is
 A) -2 B) +2/3 C) -1/2 D) +1
10. Dipole moment of HF is
 A) 1.8 B) 2.8 C) 2.0 D) 0.8
11. Which of the following is more acidic
 A) HF B) HCl C) HBr D) HI
12. Which one is strongest reducing agent
 A) HCl B) HBr C) HCl D) HI
13. HF is _____ than water
 A) More viscous B) Equally viscous C) Less viscous D) None
14. Which of the following has longest bond length
 A) HCl B) HBr C) HI D) HF
15. _____ gives slow and reversible reaction with hydrogen
 A) F B) Cl C) Br D) I
16. Chlorine acts gas _____ when it is used bleaching
 A) A dehydrating agent B) A reducing agent

- C) An oxidizing agent D) None of them
17. The order of decreasing power of F, Cl, Br and I as an oxidizing agent is
 A) $F_2 > Cl_2 > Br_2 > I_2$ B) $Cl_2 > F_2 > Br_2 > I_2$
 C) $I_2 > Br_2 > Cl_2 > F_2$ D) $Br_2 > I_2 > Cl_2 > F_2$
18. _____ directly combines with noble gases
 A) F B) Cl C) Br D) None
19. Oxidation state of _____ is always -1
 A) F B) Cl C) Br D) I
20. _____ is the formula of fluorspar
 A) Na_2AlF_6 B) CaF_2 C) $CaF_2 \cdot 3Ca_3(PO_4)_2$ D) AlF_3
21. _____ elements exist as positive ions
 A) Having lower values of E. N. B) Having high values of E.N.
 C) Intermediate values of E. N. D) All of above
22. Inter molecular forces are strongest in
 A) F_2 B) Cl_2 C) Br_2 D) I_2
23. Which has the highest I.P. value?
 A) F B) Cl C) Br D) I
24. HF is
 A) an oxidizing agent B) A reducing agent
 C) A weak acid D) A strong acid
25. Astatine belongs to group
 A) I-A B) II-A C) III-A D) VII-A
26. The only non-metal in liquid form is
 A) S B) Cl C) Br D) P
27. Smell of halogens is
 A) Fruity B) Sweet C) Irritating D) Rotten egg
28. Which compound of noble gas xenon can be prepared?
 A) Oxides B) Fluoride C) Oxyfluorides D) All of these
29. Which has complete valence shell?
 A) C B) N C) O D) Ne
30. Which is used as cooling medium in nuclear reactors?
 A) He B) Ne C) Ar D) Kr
31. Which is used in arc welding cutting?
 A) He B) Ne C) Ar D) Xe
32. Which catalyst is used in contact process?
 A) FeO_3 B) V_2O_5 C) SO_3 D) Ag_2O
33. Aqua Regia has ratio of conc. HCl and HNO_3
 A) 1 : 2 B) 3 : 1 C) 1 : 4 D) 2 : 3
34. Which gas is heavier than air?
 A) CO_2 B) NH_3 C) NO_2 and SO_2 D) All of these
35. Which is an inert gas?
 A) Nitrogen B) Oxygen C) Hydrogen D) CO

TRANSITION ELEMENTS

1. The %age of C in different types of soft iron products is in the order of
 A) Cast iron > wrought iron > steel B) Wrought iron > steel > cast iron
 C) Cast iron > steel > wrought iron D) Cast iron = steel > wrought iron
2. The d-block elements usually give _____ absorption bands
 A) Narrow B) Broad C) Normal D) None
3. Which is in different phase from other metals?

- A) Ni B) Hg C) Cd D) Na
 4. Negative ligands end with _____ in naming a complex ion
 A) 'O' B) 'ium' C) 'ate' D) 'ide'
 5. Electronic configuration of Cu^{+2} is
 A) $4\text{S}^2 . 3\text{d}^9$ B) $4\text{s}^1 . 3\text{d}^{10}$ C) $4\text{S}^0 . 3\text{d}^9$ D) None of these
 6. Haematite has chemical formula
 A) Fe_3O_4 B) Fe_2O_3 C) $\text{FeO}_3\cdot 3\text{H}_2\text{O}$ D) all
 7. $[\text{PtCl}_6]$ will form structure
 A) Square planer B) Octahedral
 C) Trigonal bipyramidal D) Tetrahedral
 8. Cl is the example of
 A) Anionic ligand B) Cationic ligand C) Neutral ligand D) None
 9. NH_3 is an example of _____ ligand
 A) Anionic B) Cationic C) Neutral D) None
 10. The anions are usually named with _____ suffix in naming a transition complex
 A) 'ium' B) 'O' C) 'ate' D) None
 11. While naming the complex compounds the cations are named always _____ anions
 A) After B) Before C) With D) in between
 12. $[\text{MnCl}_4]$ will form _____ structure
 A) Square planer B) Tetrahedral
 C) Octahedral D) Trigonal bipyramidal
 13. $[\text{Co}(\text{NH}_3)_4]^{+2}$ will form _____ structure
 A) Square planer B) Tetrahedral
 C) Octahedral D) Trigonal bipyramidal
 14. $[\text{Cu}(\text{NH}_3)_4]^{+2}$ will form _____ structure
 A) Square planer B) Tetrahedral
 C) Octahedral D) Trigonal bipyramidal
 15. Very strong paramagnetism is called
 A) Diamagnetism B) Paramagnetism C) Ferromagnetism D) none
 16. Interstitial compounds have
 A) Half formula B) Fixed formula
 C) Indefinite formula D) None
 17. The free spaces between the metal atoms and its crystal lattice are called
 A) Valances spaces B) Empty spaces
 C) Interstices D) none
 18. Any substance which have only paired electrons (with paired spins) are called
 A) Diamagnetic B) Paramagnetic C) Ferromagnetic D) none
 19. Pt and pd are used in the
 A) Hydrogenation B) Dehydrogenation
 C) Polymerization D) Oxidation of ammonia
 20. Transition metals are generally _____ reactive
 A) More B) Less C) Very much D) None
 21. The Mn^{3+} has _____ color
 A) Violet B) Green C) Red/brown D) no color
 22. There are more than _____ transition elements in the periodic table
 A) 20 B) 40 C) 60 D) 110
 23. Corrosion may be prevented by
 A) Alloying B) Paints C) Metallic coatings D) All
 24. Conversion of any metal slowly into its oxide by the action of environment is called
 A) Formation of complexes B) Corrosion
 C) Metal action D) None
 25. The electron pair donor which react with transition
 A) Electrophile B) Nucleophile C) Ligands D) none
 26. Many transition metals and their compounds are

- A) Diamagnetic B) Paramagnetic C) Neither a nor b D) none
27. Ionic and covalent compounds of T.E. are mostly
A) Coloured B) Colorless
C) Ionic coloured white covalent colorless D) Ionic colorless white covalent colored
28. Which are found abundantly in nature
A) 3d-series B) 4d-series C) 5d-series D) 4f-series
29. Zn, cd and Hg are
A) D-block elements B) f-block elements
C) both a and b D) none
30. Elements which have partially filled or f orbital in their commonly occurring oxidation states are called.
A) D-block elements B) f-block elements
C) both a and b D) transition elements
31. Which is not a bidentate ligand
A) $C_2O_4^{-2}$ B) Co_3^{-2} C) SO_4^{-2} D) None
32. The magnetic moment is denoted by
A) R B) M C) u D) None
33. Which is not a transition elements?
A) Ca B) Cu C) Cr D) Co
34. The colour of transition metal complex is due to
A) d-d transitions of electrons
B) Paramagnetic nature of transition elements
C) Ionizaiton
D) Loss of s-electrons
35. The total number of transition elements is more than
A) 10 B) 14 C) 40 D) 50
36. Rusting can be avoided by
A) Making alloys B) Tin or Zinc plating C) Use of enamel D) All of these
37. Content of carbon in steel is
A) 0.12 to 0.25% B) 0.25 to 2.5% C) 3.0 to 3.5% D) 4.0 to 4.5%
38. Coinage metal is
A) Co B) Ag C) Pt D) Ni
39. Transition elements have binding energies
A) Low B) High C) No effect D) All of these
40. Greater number of unpaired electrons stronger will be the
A) Banding energies B) Melting and boiling points
C) Paramagnetism D) All of these
41. What type of a coordination sphere may be
A) Anionic B) Cationic C) Neural D) All of these
42. Formula of limonite is
A) le_2O_3 B) Fe_3O_4 C) $Fe_2O_3.3H_2O$ D) FeS_2
43. $KMnO_4$ has colour
A) Orange red B) Purple C) Yellow D) Blue
44. Manganese shows oxidation number in $KMnO_4$
A) I B) V C) VI D) VII

FUNDAMENTAL PRINCIPLES OF ORGANIC CHEMISTRY

1. For geometric isomerism the two groups attached to the same carbon atom must be
A) Similar B) Dissimilar C) May be a or b D) none of them
2. If dissimilar groups are attached on one side it is called
A) Cis-form B) Trans-form C) Tautomeric form D) none of them
3. If similar groups are attached on one side it is called
A) Cis form B) Trans-form C) Tautomeric form D) none of them
4. Diethyl ketone methyl n-propyl ketone are an example of
A) Chain isomerism B) Positional isomerism

- C) Metamerism D) Functional group isomerism
5. 1-butene and 2-butene are an example of
A) Chain isomerism B) Positional isomerism
C) Metamerism D) Functional group isomerism
 6. n-butane and iso butane are an example of
A) Chain isomerism B) Positional isomerism
C) Metamerism D) Functional group isomerism
 7. Alkynes normally have _____ hybridization
A) sp B) sp^2 C) sp^3 D) $d\ sp^3$
 8. Alkenes normally have _____ hybridization
A) sp B) sp^2 C) sp^3 D) $d\ sp^3$
 9. Alkanes normally have _____ hybridization
A) sp B) sp^2 C) sp^3 D) $d\ sp^3$
 10. The active part in a molecule is called
A) Heteroatom B) Functional group C) Chemical bond D) Ion
 11. The atom other than C in a heterocyclic compound is called as
A) Hetero atom B) Hetero ion C) Hetero molecule D) Hetero element
 12. Heterocyclic compounds are also called as
A) Carbocyclic B) Heterocycles C) Heteroatom D) None of them
 13. Those compound which have any atom other than C as member of rings are called as
A) Monocyclic B) Heterocyclic C) Aliphatic D) Both a and b
 14. Closed chain compounds can be classified into
A) Homocyclic B) Heterocyclic C) Aliphatic D) Both a and b
 15. The open chain compounds are also called
A) Aliphatic B) Alicyclic C) Aromatic D) Both a and b
 16. Which is a knocking inhibitor
A) TEL B) CH_4 C) C_2H_3 D) C_6H_6
 17. The gasoline having octane No. 100 has _____
A) More knocking B) Less knocking C) No knocking D) Only knocking
 18. The gasoline with high octane No. has _____
A) More knocking B) Less knocking C) No knocking D) Only knocking
 19. The by-product of cracking can be
A) Ethene B) Butene C) Benzene D) All of them
 20. Cracking done at low pressure and with a catalyst is called _____ cracking
A) Thermal B) Catalytic C) Steam D) None of them
 21. Cracking done with heat at high temperature is called as _____ cracking
A) Thermal B) Catalytic C) Steam D) None of them
 22. Cracking normally gives smaller _____
A) Alkanes B) Alkenes C) Alkynes D) Both a and b
 23. Fractional distillation of petroleum gives gasoline
A) 10% B) 20% C) 30% D) 40%
 24. Petroleum when as black substance is called as
A) Crude oil B) Crude solid C) Mineral oil D) Mineral liquid
 25. Petroleum in its refined form is also called as
A) Mineral oil B) Mineral gas C) Mineral solid D) Mineral gas Mineral oil
 26. Natural gas in Pakistan is used as
A) Automobile fuel B) Industrial fuel C) Domestic fuel D) All of them
 27. Which is a mixture of low boiling hydrocarbon
A) Natural gas B) Petroleum C) Coal D) Wood
 28. Due to the bacterial action on wood it is converted into
A) Peat B) Lignite C) Bituminous D) Anthracite
 29. Which is not a fossil fuel
A) Petroleum B) Coal C) Natural gas D) none of them
 30. Which is not an organic compound

- A) Rubber B) Paper C) Dyes D) None of them
 31. Organic compounds are generally react at _____ solvents
 A) Polar B) Non-polar C) Partially polar D) None of them
 32. Organic compounds generally react at _____ rates
 A) Slow B) Fast C) Moderate D) None of them
 33. Which of the following is complex
 A) Glucose B) $(C_6H_{10}O_5)_n$ C) Fructose D) Sugar
 34. Similarly in properties of different organic compounds give rise to the understanding of
 A) Polymerization B) Non-polar nature C) Homologous D) Isomerism
 35. To which carbon is not multiply bonded in organic compounds
 A) Nitrogen B) Sulphur C) Oxygen D) Hydrogen
 36. The self linking property of carbon is called as
 A) Linking polymerization B) Addition
 C) Catenation D) Elimination
 37. Ethers show the phenomenon of
 A) Position isomerism B) Functional group
 C) Metamerism D) Cis-trans isomerism
 38. Tetraethyl lead causes disease
 A) Typhoid B) Respiratory C) Stomach D) Muscular
 39. General formula of carboxylic acids is
 A) RCOH B) RCOR C) RCOOH D) RO-H
 40. An element essential for organic compounds.
 A) Hydrogen B) Carbon C) Nitrogen D) Halogen
 41. The saturated alicyclic hydrocarbons have the general formula
 A) C_nH_{2n+2} B) C_nH_{2n+1} C) C_nH_{2n} D) C_nH_{2n-1}
 42. Functional group or nitriles is
 A) $-NH_2$ B) $\begin{array}{c} \diagup \\ C \\ \diagdown \end{array} = NH$ C) $\begin{array}{c} \parallel \\ -C-NH_2 \end{array}$ D) $-C \equiv N$
 43. The % age character of 's' and 'p' in sp^3 is respectively
 A) 25% & 75% B) 33% & 66% C) 50% & 50% D) 30% & 70%
 44. 2-butene shows isomerism
 A) Structural B) Cis-trans C) Metamerism D) Functional group

ALIPHATIC HYDRO CARBONS

1. Alkynes having C_4 to C_{12} chains are _____ in nature
 A) Liquids B) Solids C) Gases D) Plasma
 2. First three members of alkyne series are
 A) Liquids B) Solids C) Gases D) Plasma
 3. Dehalogenation of tetra halides happens in the presence of active metal like
 A) Zn B) Mg C) Both a and b D) None of them
 4. Markovnikov's rule is applicable to _____ alkenes
 A) Symmetrical B) Unsymmetrical C) Both a and b D) None of them
 5. The heat of hydrogenation of most of alkenes is about _____ KJ/mole:
 A) 1000 B) 120 C) 140 D) 103
 6. The dehalogenation of vicinal dihalides involves Zn and:
 A) Anhydrous solvent B) Methanol C) Acetic acid D) Any of these
 7. The order of dehydration of alcohol:
 A) $1^0 > 2^0 > 3^0$ B) $1^0 > 3^0 > 2^0$ C) $2^0 > 3^0 > 1^0$ D) $3^0 > 2^0 > 1^0$
 8. Which can be used for dehydration of alcohol:
 A) P_4O_{10} B) H_2SO_4 C) H_3PO_4 D) All of them
 9. Alkyl halide can be dehydrohalogenated to alkene in the presence of:
 A) KOH B) KOH (aq.) C) KOH (alc.) D) All of them
 10. Alkene having only one double bond are called:
 A) Monoene B) Diene C) Dienophile D) None of them
 11. Which are the steps of free radical mechanism:
 A) Initiation B) Propagation C) Termination D) All of them
 12. The correct order of reactivity of halogens with alkanes is

13. The complete combustion of alkanes normally gives
 A) $I_2 > Br_2 > Cl_2 > F_2$ B) $I_2 > Cl_2 > F_2 > Br$ C) $F_2 > Cl_2 > I_2 > Br_2$ D) $F_2 > Cl_2 > Br_2 > I_2$
 A) $CO_2 + H_2O$ B) $CO + H_2O$ C) CO_2 only D) CO only
14. The general reaction of alkanes are based on mechanism:
 A) Free radical B) Ion exchange C) Ionic D) None of them
15. The B.P. of branched hydrocarbons are _____ than their normal chain isomers
 A) Less B) More C) Equal D) None
16. The B. P. increases by _____ for addition of CH_2 group to an alkane molecule
 A) $20^\circ C$ B) $40^\circ C$ C) $30 \text{ ----- } 40^\circ C$ D) $20 \text{ ----- } 30^\circ C$
17. Which can dissolve alkanes:
 A) Benzene B) n-hexane C) Ether D) All of these
18. Alkane are normally
 A) Nonpolar B) Polar C) Strongly polar D) All of these
19. C_{18} and onwards hydrocarbons are normally
 A) Gases B) Liquids C) Solids D) Plasma
20. C_5 _____ C_{17} hydrocarbons are normally
 A) Gases B) Liquids C) Solids D) Plasma
21. C_1 _____ C_4 hydrocarbons are normally
 A) Gases B) Liquids C) Solids D) Plasma
22. Which is a free radical?
 A) CH_4 B) CO_2 C) CH_2COO D) CH_3
23. To prepare alkene from alkyl halide the process used is:
 A) Substitution B) Hydrogenolysis C) Elimination D) none of them
24. The temp used for the hydrogenation of alkene using Ni is
 A) $200^\circ C$ B) $400^\circ C$ C) $300^\circ C$ D) $1000^\circ C$
25. Alkenes are hydrogenated to alkanes in the presence of
 A) Ni B) Pt C) Both a and b D) none of them
26. Alkenes normally have _____ geometry
 A) Tetrahedral B) Linear C) Planer D) None
27. Alkenes are also named by replacing ene with
 A) Ane B) Ylene C) Yne D) Die
28. To different isomers we use
 A) n---- B) iso---- C) neo D) all of them
29. The hydrocarbons having triple bonds normally end with suffix
 A) Ane B) Ene C) Yne D) Oic
30. The hydrocarbons having double bonds normally end with suffix
 A) Ane B) Ene C) Yne D) Oic
31. The saturated hydrocarbons usually end with suffix _____
 A) Ane B) Ene C) Yne D) Oic
32. If an aliphatic hydrocarbon does not have its valences fully satisfied bit it has double bonds it is called as _____ hydrocarbons
 A) Carboxylic B) Saturated C) Unsaturated D) Aromatic
33. If an aliphatic hydrocarbon has its valences fully satisfied is called as _____ hydrocarbons
 A) Carboxylic B) Saturated C) Unsaturated D) Aromatic
34. Grignard reagent is reactive due to
 A) The presence of halogen atom B) The presence of Mg atom
 C) The polarity of C – Mg bond D) None of these
35. Wurtz reaction is used to prepare
 A) Alcohol B) Alkyl halide C) Ether D) Alkanes
36. $R - Mg - X$ reagent was prepared by
 A) Victor Grignard B) Frankland C) Wurtz D) Kolbe
37. Reaction of Grignard's reagent with ethylene epoxide gives
 A) 1-butanol B) 2-butanol C) Butane D) Ethanol
38. Which one gives acidic reactions?
 A) $CH_2 - C \equiv C - CH_2$ B) $CH_2 - CH_2 - HC = CH - CH_3$
 C) $CH_2 - CH = CH_2$ D) $CH_2 - C \equiv CH$
39. Removal of hydrogen from alkane is called
 A) Dehydrogenation B) Alkanes C) Dehalogenation D) Dehydrohalogenation

40. Simplest member of alkanes is
A) Methane B) Ethane C) Propane D) Butane
41. Alkanes are inert at ordinary conditions towards
A) Acids B) Alkalis C) Oxidizing agents D) All
42. Formation of carbon black and CO from alkanes is called
A) Nitration B) Incomplete oxidation C) Catalytic oxidation D) Combustion
43. Higher fatty acids of alkanes are prepared from the reaction of alkanes with
A) Nitration B) Oxidation C) Catalytic oxidation D) Combustion
44. Conversion of alkane into alcohols is called
A) Nitration B) Oxidation C) Catalytic oxidation D) Combustion
45. Which is used as solvent?
A) Propane B) Alkanes C) Nitroalkanes D) Alkane sulphonic acid
46. Dehydrating reagents is
A) Al_2O_3 B) P_4O_{10} C) Conc. H_2SO_4 D) All

AROMATIC HYDRO CARBONS

1. The electron density is present _____ of the benzene ring
A) Side wise B) Above C) Below D) Both b and c
2. The molecular mass determined by vapour density method for benzene was
A) 70. 108 B) 80. 108 C) 90. 108 D) 78. 108
3. Atomic orbital treatment gave the concept of
A) Hybridization B) Localization C) Delocalization D) none of these
4. Benzene the un hybridized orbital is
A) p_x B) p_y C) p_z D) sp^3
5. Benzene normally show _____ reactions
A) Addition B) Substitution C) Limiting D) Elimination
6. The structure of benzene is
A) Planer B) Tetrahedral C) Linear D) Coplanar
7. For halogenations of benzene normally we use
A) Light B) Radiation C) Sunlight D) none of these
8. The atom or group attaches opposite to para position is called as
A) Reference group B) Main group C) Reactive group D) Functional group
9. The substituent which is attached first to the ring before other substituents is called _____
A) Reference group B) Main group C) Reactive group D) Functional group
10. A single benzene ring can have _____ para positions maximally
A) One B) Two C) Three D) Four
11. A single benzene ring can have _____ ortho positions maximally
A) One B) Two C) Three D) Four
12. For sulphonation of benzene with H_2SO_4 the temperature is _____
A) 50 B) 80 C) 25 D) 100
13. For sulphonation of benzene with SO_3 the temperature is 0°C
A) 50 B) 80 C) 25 D) 100
14. The temp. for nitration of benzene is _____
A) 50-55 B) 100 C) 170 D) 150
15. Nitration of benzene happens by mixing HNO_3 and H_2SO_4 in _____ ratio
A) 1 : 2 B) 1 : 4 C) 1 : 3 D) 1 : 1
16. For halogenations the poorest yield is given by
A) Chlorination B) Bromination C) Iodination D) Fluorination
17. For halogenations reactions which is the fastest reaction
A) Chlorination B) Bromination C) Iodination D) Fluorination
18. Benzene can react with _____ easily to produce substituted product
A) Nucleophile B) Electrophiles
C) Weak Electrophile D) Strong Electrophiles
19. Which is called soda lime
A) NaOH B) CaO C) NaOH + CaO D) Na_2CO_3
20. Cyclic polymerization of _____ in the presence of organonickel yields benzene
A) Ethene B) Ethylene C) Ethane D) Ethyne

21. Cyclohexane can be converted not benzene in the presence of
A) Pt at 100°C B) Pt at 250°C
C) Pd at room temperature D) Pt at room temperature
22. Dewar predicted _____ closed chain structure for benzene
A) Two B) Three C) Seven D) Five
23. For benzene _____ closed chain structure were proposed
A) Two B) Five C) Seven D) No
24. For benzene _____ open chain structure were proposed
A) Two B) Five C) Seven D) No
25. Which is not a representation of benzene
A) Ph B) Aryl C) Phenyl D) none of them
26. Which of the following is not a polycyclic aromatic hydrocarbon?
A) Biphenyl B) Naphthalene C) Naphthol D) None of them
27. Which of the following is a monocyclic aromatic hydrocarbon
A) Toluene B) Phenol C) Aniline D) All of them
28. The simplest and the parent members of aromatic hydrocarbon is
A) Benzene B) Toluene C) Biphenyl D) Naphthalene
29. Which of the following generally have low carbon to hydrogen ratio
A) Alkanes B) Alkynes C) Alkenes D) Aromatic hydrocarbon
30. The electrophile in aromatic Sulphonation is
A) H_2SO_4 B) HSO_4 C) SO_3 D) HSO_3
31. Benzene gives reactions generally
A) Electrophilic substitution B) Addition
C) Synthesis D) Addition and electrophilic substitution
32. Oxidation of benzene occurs in the presence of catalyst
A) V_2O_5 B) Ni C) Al_2O_3 D) FeCl_3
33. Empirical formula mass of benzene is _____ times lesser than molecular formula mass
A) Four B) Five C) Six D) Seven
34. Arenes are also called
A) Atom B) Hydrocarbons C) Aromatic D) Benzene
35. Aromatic of higher molecular mass are degraded and generated basic fundamental unit of molecule of?
A) Benzene B) Toluene C) Aniline D) Biphenyl
36. Biphenyl has benzene rings
A) One B) Two C) Two condensed rings D) Two isolated rings

ALKYL HALIDES

1. _____ is called a synthetic tool in the hand of a chemist
A) Ethyl alcohol B) Sulfuric acid C) Benzene D) Grignard's reagent
2. In general the reactions of Grignard's reagents are carried out in
A) Acetone B) Ether C) Pyridine D) n-hexane
3. _____ is produced by the reaction of ethyl chloride with alc – KOH
A) Ethanol B) Ethene C) Ether D) Acetylene
4. The reaction in which an atom or group is replaced by another atom or group is called
A) Substitution B) Elimination C) Both a and b D) None
5. When carbon dioxide is passed through the R-Mg-X is produced
A) Any carboxylic acid B) Propanoic acid C) Propanedioic acid D) None of these
6. By reaction of Grignard's reagent with the ammonia we get
A) Alkanes B) Ethane C) Methane D) None of these
7. The best way to get alkanes from the Grignard's reagent is reacting with
A) Benzene B) R-X C) Water D) None of these
8. By reacting Grignard's reagent with the HCHO we get
A) 1° -alcohol B) 2° -alcohol C) 3° -alcohol D) All of these
9. Grignard's reagent is
A) An organic compound B) An organo metallic compound
C) An inorganic compound D) None
10. Elimination reaction is
A) E_1 B) E_2 C) Both a and b D) None of these

11. The transition state is formed in the
A) S_N1 B) S_N2 C) Both a and b D) None of these
12. S_N2 reaction is _____ step reaction
A) Uni B) Bi C) Multi D) All of these
13. The S_N1 reaction is generally given by
A) $1^\circ R \text{ --- } X$ B) $2^\circ R \text{ --- } X$ C) $3^\circ R \text{ --- } X$ D) All of these
14. Nucleophile generally attack the site of molecule which is
A) Electron rich B) Electron deficient C) Bonded D) All of these
15. The elimination of elements of H and X from two adjacent C-atoms is called
A) Dehalogenation B) Dehydrohalogenation C) Dehydrogenation D) All
16. In Wurtz's reaction the metal used is
A) Zn B) Ag C) Na D) None
17. The formula of thionyl chloride is
A) $COCl_2$ B) $SOCl$ C) $SOCl_2$ D) None
18. Monohalo derivative of alkanes are called
A) Chloroalkanes B) Alkyl halides C) Aromatic halides D) None of these
19. When Grignard's reagent react with acetone it gives
A) 1° ----- alcohol B) 2° ----- alcohol C) 3° ----- alcohol D) All of these
20. Which is more reactive
A) $R \text{ --- } I$ B) $R \text{ --- } Br$ C) $R \text{ --- } Cl$ D) $R \text{ --- } OH$
21. When $R \text{ --- } OH$ react with $H \text{ --- } X$ then the product will be
A) $R \text{ --- } X$ B) Water C) Both a and b D) None
22. The rate determining step in the S_N1 reaction is
A) Carbonium ion formation B) Product formation
C) Nucleophilic attack D) None
23. The order of stability 1° , 2° and 3° alkyl halides is
A) $1^\circ > 2^\circ > 3^\circ$ B) $2^\circ > 1^\circ > 3^\circ$ C) $3^\circ > 1^\circ > 2^\circ$ D) $3^\circ > 2^\circ > 1^\circ$
24. The general formula of alkyl halides is
A) $C_nH_{2n+1}X$ B) $C_nH_{2n-1}X$ C) $C_nH_{2n+1}X$ D) $C_nH_{2n-2}X$
25. Generally the reaction of alkyl halides fall into
A) Substitution B) Elimination C) Both a and b D) None of these
26. Both E_1 and E_2 mechanism can be shown by
A) $1^\circ \text{ --- } RX$ B) $2^\circ \text{ --- } RX$ C) $3^\circ \text{ --- } RX$ D) None of these
27. E_1 mechanism is generally shown by
A) $1^\circ \text{ --- } RX$ B) $2^\circ \text{ --- } RX$ C) $3^\circ \text{ --- } RX$ D) None of these
28. Primary alkyl halides follow
A) E_1 mechanism B) E_2 mechanism C) Both E_1 and E_2 D) None of these
29. Elimination reactions are usually _____ catalyzed
A) Acid B) Base C) PI D) NI
30. In $R \text{ --- } X$ the 'R' cannot be
A) Alkane B) Saturated hydrocarbon C) Aromatic D) Aliphatic
31. The alkyl halide molecule on which a nucleophile attacks is called
A) Substrate B) Substituent C) Substituted D) All of these
32. A nucleophile may be
A) Positive B) Negative C) Neutral D) Both band c
33. An electrophile may be
A) Positive B) Negative C) Neutral D) Both c and a
34. Which is a good nucleophile as well as a good leaving group
A) F B) Cl C) Br D) I
35. Which is a weak nucleophile
A) OH B) Br C) NH_3 D) Cl
36. Which is not a nucleophile
A) OH B) C_2H_5O C) SCN D) None of these
37. Which is the major failure towards the reactivity of alkyl halide
A) Bond energy B) Bond polarity C) Bonding time D) None of these
38. With the increase in size of halogen atom the reactivity of an alkyl halide
A) Increase B) Decrease C) Remain constant D) None of these
39. Which bond is most stable?
A) $C \text{ --- } Cl$ B) $C \text{ --- } F$ C) $C \text{ --- } Br$ D) $C \text{ --- } I$

40. Best method of preparation of alkyl halide from alcohols is by its reaction with
A) HX B) SOCl_2 C) PX_5 and PX_3 D) All
41. E_2 has molecularity
A) One B) Two C) Three D) Half
42. Hydrolysis of Grignaro's reagent gives
A) Alkane B) Alkyl halide C) Alcohol D) Carboxylic acid
43. An alkyl halide with two halogen atoms at same carbon is called
A) Vic-dihalide B) Gem-dihalide C) Poly haloalkane D) Primary alkyl halide
44. Formula of methylene chloride is
A) CH_2Cl_2 B) CH_3Cl C) CHCl_3 D) CCl_4
45. Reaction of PCl_5 with ethanol gives
A) Ethyl chloride B) Ethyl chloride & POCl_3
C) HCl D) Ethyl chloride & POCl_3 & HCl
46. The strength of R – F bond is
A) Weak B) Weaker C) The Weakest D) Strongest
47. Which property is not present in SN_1 reaction?
A) Unimolecular B) Order of reaction two
C) No transition state D) Two steps mechanism
48. Which step does not occur in SN_1 reactions?
A) Inversion B) Ionization
C) Attack and removal of nucleophiles in different steps D) Unimolecular
49. Which group has least priority?
A) $-\text{COOH}$ B) $-\text{CN}$ C) $-\text{NH}_2$ D) $-\text{R}$
50. Benzene was discovered by
A) Michael Faraday B) Lavoiser C) Hoffmann D) Proust
51. Empirical formula of benzene is
A) C_6H_6 B) C_4H_4 C) C_3H_3 D) C

ALCOHOLS, PHENOLS AND ETHERS

1. Which is most acidic
A) Methanol B) Ethanol C) Carboic acid D) Propanol
2. Which is called king of organic solvents?
A) Ethers B) Esters C) Alcohol D) None of these
3. Those ethers which have two same alkyl groups are called _____ ether
A) Simple B) Complex C) Mixed D) None of these
4. The reaction of phenol with NaOH is an example of reaction of _____ group
A) OH B) Ring C) Phenyl D) None of these
5. The pH of phenolic solution is normally
A) 3 B) 2 C) 5 D) 6.5
6. Phenol is _____ soluble in water
A) Completely B) Partially C) Not at all D) None of these
7. Which alkali is used in Dow's method
A) NaOH B) KOH C) Na_2CO_3 D) RbOH
8. Carboic acid is the other name for
A) Methanol B) Ethanol C) Propanol D) Phenol
9. In Lucas test primary alcohol make an oily layer
A) Immediately B) In 10 mins C) On heating D) Not
10. In Lucas test secondary alcohol make an oily layer
A) Immediately B) In 10 mins C) On heating D) Not
11. In Lucas test tertiary alcohol make an oily layer
A) Immediately B) In 10 mins C) On heating D) Not
12. Which is a reactant for Lucas test
A) $\text{ZnCl}_2 + \text{ConcHCl}$ B) Zn C) Na D) Br
13. Dehydration of an alcohol at 140° and conc. H_2CO_4 gives
A) Alkene B) Ether C) Ester D) Any hydride
14. Dehydration of an alcohol at 180°
A) Alkene B) Ether C) Ester D) Any hydride
15. Which as an example is O ---- H bond reactivity of an alcohol when it reacts with

- A) HCl B) NH₃ C) SOCl₂ D) Na
16. The order of reactivity of alcohols when O ---- H bond breaks
 A) Primary > secondary > tertiary B) Tertiary > primary > secondary
 C) Secondary > tertiary > primary D) Tertiary > Secondary > primary
 17. The order of reactivity of alcohols when C ---- O bond breaks
 A) Primary > secondary > tertiary B) Tertiary > primary > secondary
 C) Secondary > tertiary > primary D) Tertiary > Secondary > primary
 18. If an electrophile attacks then _____ bond breaks first
 A) C ---- H B) O ---- H C) C ---- O D) None of these
 19. If a nucleophile attacks on alcohol then _____ bond breaks first
 A) C ---- H B) O ---- H C) C ---- O D) None of these
 20. The acidic reactivity of an alcohol is due to _____ bond
 A) C ---- H B) O ---- H C) C ---- O D) None of these
 21. Methyl alcohol and ethyl alcohols are
 A) Liquids B) Solids C) Gases D) Plasma
 22. With the increase in carbon number the solubility of an alcohol
 A) Increases B) Decreases
 C) Remains unaffected D) None of these
 23. Lower alcohols are generally
 A) Colourless B) Coloured C) Highly viscous D) None of these
 24. Alcohol containing 10% methyl alcohol in it is called
 A) Rectified spirit B) Spirit of wine C) Spirit D) Mehtylated spirit
 25. 95% alcohol is called
 A) Rectified spirit B) Spirit of wine C) Spirit D) Mehtylated spirit
 26. Alcohol obtained by fermentation is up to
 A) 10% B) 12%-14% C) 20% D) 35%
 27. The pressure used to prepare methanol from water gas is
 A) 1 atm B) 100 atm C) 200 atm D) 300 atm
 28. To prepare ethanol by fermentation the optimum temp. is
 A) 10----20°C B) 25----25°C C) 20°C D) 35°C
 29. Fermentation is a _____ precess
 A) Biological B) Chemical C) Biochemical D) Physical
 30. Temperature used for preparing methanol form water gas is
 A) 4000°C B) 450°C C) 1000°C D) 1200°C
 31. New a days Methanol is prepared from
 A) CO B) Co₂ C) H₂ D) Water gas
 32. Wood spirit is the other name for
 A) Methanol B) Ethanol C) Butanol D) Benzene
 33. Formerly methanol was prepared by
 A) Wood B) Carbon C) Wine D) All of these
 34. Alcohol may be
 A) Saturated B) Unsaturated C) Both a and b D) None
 35. Which of the following has minimum hydrogen bonding
 A) Water B) Alcohol C) Ether D) Phenol
 36. The nature of alcohol (i.e. primary, secondary and tertiary) depends upon nature of _____ group
 A) R B) OH C) Both a and b D) None of these
 37. Which of the following is an example of trihydric alcohol?
 A) Methanol B) Glycol C) Glycerol D) None of these
 38. Which of the following are considered as derivatives of water
 A) Alcohol B) Phenol C) Ether D) All of these

MACROMOLECULES

1. A nucleoside may be

- A) Ribonucleoside B) Deoxyribonucleoside
C) Both a and b D) None
2. When a nitrogenous base combine with a sugar it is called
A) Ribose B) Nucleosides C) Nucleosides D) None
 3. Purines include
A) Adenine B) Guanine C) Both A and B D) None of them
 4. RNA has _____ instead of _____ in comparison to DNA
A) T and C B) U and T C) T and U D) C and T
 5. Oxidation reduction is done by
A) Oxidoreductase B) Lipases C) Lyase D) None of these
 6. Which is not an enzyme
A) Transverses B) Lipase C) Lyase D) None of these
 7. Which is not a steroid
A) Transverses B) Lipase C) Lyase D) None of these
 8. Naturally occurring lipids are called
A) Fats B) Protein C) Steroids D) None
 9. Sterols, vitamin D and terpenes belong to
A) Simple lipids B) Complex lipids C) Derived lipids D) None
 10. Ester of fatty acids with glycerol are
A) Primary B) Secondary C) Tertiary D) None
 11. The three dimensional folding and twisting of a polypeptide chain give rise to
A) Primary structure B) Secondary structure
C) Tertiary structure D) All of these tertiary
 12. Protein may have
A) Primary structure B) Secondary structure
C) Tertiary structure D) All of these
 13. The proteins which are derived by conjugated proteins are called as
A) Simple protein B) Complex protein C) Derived protein D) None
 14. When a protein is attached with a non-protein group that non-protein group is called
A) Activator B) Macro-protein C) Prosthetic D) None
 15. The protein which only yield amino acids and their derivative
A) Simple proteins B) Complex proteins C) Derived protein D) None
 16. Protein are classified into
A) Simple protein B) Complex proteins C) Derive proteins D) All of these
 17. A high molecular weight organic materials, which upon hydrolysis give amino acid is called.
A) Fats B) Protein C) Carbohydrates D) None of these
 18. β -D-glucose is a monomer for
A) Starch B) Cellulose C) Glycogen D) Protein
 19. α -D-glucose is a monomer for
A) Starch B) Cellulose C) Glycogen D) Protein
 20. Cotton is _____ % cellulose
A) 90 B) 100 C) 99 D) 30
 21. Glycogen is stored in
A) Animals B) Plants C) Soil D) None of these
 22. Polysaccharides are also called
A) Crystals B) Sugars C) Liquids D) Non-sugars
 23. Monosaccharide and oligosaccharides are generally called as
A) Crystals B) Sugars C) Liquids D) Non-sugars
 24. Monosaccharides are
A) Aldoses B) Ketoses C) Either a and b D) None of these
 25. $(CH_2O)_n$ is general formula for
A) Monosaccharides B) Oligosaccharides C) Polysaccharides D) None of these
 26. The general formula for carbohydrate is
A) $C_x(H_2O)_y$ B) C_nH_{2n+2} C) CH D) None of these

27. The temp. and pressure used for PVC polymerization is
A) 10°C and 10 atm B) 20°C and 20 atm C) 52°C and 9 atm D) 80°C and 10 atm
28. Which polymerization is free radical mechanism based
A) Addition B) Condensation C) Both a and b D) None
29. The polymers which can not be re-softened again and again are called
A) Thermoplastic B) Thermosetting C) Both a and b D) None
30. The polymers which can be re-softened again and again are called
A) Thermoplastic B) Thermosetting C) Both a and b D) None
31. A polymer may be
A) Homopolymer B) Co-Polymer C) Terpolymer D) All of these
32. Mostly the molecular mass of macro molecules ranges as between 10,000 to
A) One lac B) One million C) One billion D) One trillion
33. Molecular mass of polymer = Mol. Mass of monomer X _____
A) 1000 B) 1 C) D.P. D) All of these
34. In macromolecules DP stands for
A) Dissociation Parameter B) Dissociation polymer
C) Degree of polymerization D) None of these
35. A polymer may be
A) Linear B) Branches C) Cross linked D) A of these
36. The other name for cross linked polymers is
A) Linear polymer B) Branched polymers
C) Inter connected polymers D) None of these
37. Which is not a Biopolymer
A) Lipids B) Protein C) Carbohydrate D) None of these
38. Organic macromolecules are classified into
A) Biopolymers B) Synthetic polymers
C) Both a and b D) None
39. The large molecules which are formed by the monomers called as
A) Micro molecules B) Macro molecules C) Both a and b D) None
40. Macromolecules are classified into
A) Organic B) Inorganic C) Both a and b D) None

COMMON CHEMICAL INDUSTRIES IN PAKISTAN

1. In cement rotary kiln decomposition zone has a temperature of
A) 500°C B) 900°C C) 1500°C D) None of these
2. In cement rotary kiln pre heating zone has a temperature of
A) 500°C B) 800°C C) 1500°C D) None of these
3. The slurry is heated in rotary kiln for _____ hours
A) 1 – 2 B) 2 – 3 C) 3 – 4 D) 5 – 6
4. The slurry in wet process have _____ % water
A) 10 – 20% B) 20 – 30% C) 35 – 45% D) 50%
5. The ratio of limestone and clay in wet process is
A) 1 : 1 B) 2 : 1 C) 3 : 1 D) 4 : 1
6. In Pakistan most successful process of cement making is _____ process
A) Dry B) Wet C) Long D) Short
7. The major constituent of cement is
A) Lime B) Silica C) Alumina D) Magnesia
8. A mixture of lime, silica, iron oxide and aluminium is called
A) Dry B) Wet C) Portland D) None of these
9. The annual production of Urea in Pakistan is almost
A) 5.6 million metric tons B) 10.2 million metric tons
C) Calcium fertilizer D) Potassium fertilizer
10. KNO₃ fertilizer has _____ % nitrogen
A) 10 B) 13 C) 18 D) 46

11. Which is given at later stages of growth
A) Nitrogen B) Phosphorous fertilize
C) Calcium fertilize D) Potassium fertilize
12. Ammonium nitrite is used in combination with
A) Limo B) Lime stone C) Quick lime D) All of these
13. Ammonium carbamate is produced by mixing CO_2 and NH_3 in _____ ratio
A) 1 : 1 B) 1 : 2 C) 1 : 3 D) 2 : 1
14. Most concentrated solid nitrogen fertilizer is
A) Ammonia B) Nitrate C) Urea D) Calcium super nitrate
15. Acidity of soil is controlled by adding
A) Lime B) NaOH C) KOH D) HCl
16. All nitrogen fertilizers make the soil acidic except
A) Calcium nitrate B) Sodium nitrate C) Potassium nitrate D) All of these
17. A good fertilizer
A) Never changes P^{H} of the soil B) Is soluble in water
C) Is not injurious to plants D) All of these
18. Macronutrient required per acre is from 5 kg to
A) 200 kg B) 500 kg C) 700 kg D) 100 kg
19. Which is not a macronutrient?
A) N B) C C) S D) None of these
20. The micronutrients requires per acre are from 6 gms
A) 10 gms B) 20 gms C) 200 gms D) 400 gms
21. Micronutrients are _____ in excess
A) Essential B) Essential C) Poisonous D) Required
22. The nutrient which are required in small amount for the growth of a plant are called
A) Micronutrient B) Macronutrient C) Trace element D) Both A and B
23. The fertilizers consisting of faeces and urine of domestic livestock is called
A) Plan fertilizers B) Synthetic fertilizers C) Manure D) No

HYDROGEN

1. Which of the following forms polymeric hydrides
A) Na B) K C) F D) Be
2. NaBH_4 and LiAlH_4 are
A) Ionic hydrides B) Covalent hydride C) Interposol hydrides D) Complex hydrides
3. The elements which does not make salt like hydride is
A) Na B) Ca C) K D) Si
4. Which is not interstitial hydride?
A) LaH B) VH C) TaH D) None
5. Ionic hydrides are generally
A) Liquid at room temperature B) Good electrical conductors
C) Good reducing agents D) Easily reduced
6. The structure of complex hydrides is
A) Tetrahedral B) Trigonal C) Octahedral D) Square planer
7. The laws of chemical combination are not obeyed by
A) Ionic hydride B) Increases C) Complex hydride D) Interstitial hydride
8. NaH is
A) Ionic hydride B) Complex hydride C) Covalent hydride D) Interstitial hydride
9. The valence shell of hydrogen is half filled like chose of
A) IV ---- A B) VIA C) V ---- A D) VIIA
10. The atomic weight of hydrogen is
A) 1.0 B) 1.08 C) 1.008 D) 1.80
11. The binary compounds of halogens are called
A) Hydrides B) Halides C) Oxides D) Nitrides
12. Hydride with ionic bond is called
A) Ionic B) Covalent C) Only A & B D) None
13. Hydride of Be and Mg form type of halides
A) Ionic B) Covalent C) Polymeric D) A, B & C

14. Hydrogen bond is the strongest between the molecules of?
A) HF B) HCl C) HBr D) HI
15. Hydrides are classified based upon
A) Bonding B) Physical states C) m.p D) Nature
16. Reaction of hydrogen with group I.A and II-A give hydrides
A) Ionic B) Covalent C) Intermediate D) Complex
17. Hydrides of Zn & Cd are in nature
A) Ionic B) Covalent C) Intermediate D) Complex
18. Reaction of hydrides of group I-A & III-A give type of hydrides
A) Ionic B) Covalent C) Intermediate D) Complex
19. Adsorption of hydrogen in transition elements give hydrides
A) Ionic B) Interstitial C) Intermediate D) Complex
20. Hydrides used as catalysts generally are
A) Ionic B) Covalent C) Intermediate D) Complex
21. Tendency of hydrides to be covalent in a period from left to right
A) Decreases B) Remains same C) Increase D) All
22. The non-conductor, usually gaseous and volatile Hydrides are?
A) Ionic B) Covalent C) Intermediate D) Complex
23. Stability of hydrides in a group from top to bottom
A) Decreases B) Remains same
C) Increases D) May increases or decreases
24. Most stable hydrides is of
A) Fluorine B) Bismuth C) Thallium D) Lead
25. Hydrides form hydrogen bonding
A) H₂O B) NH₃ C) HF D) All of these
26. Boiling point of hydride is the greatest for
A) H₂O B) H₂S C) NH₃ D) CH₄
27. Hydrides formed between elements with E. N difference greater than 1.8 is
A) Ionic B) Covalent C)Complex D) Intermediate

ANSWER KEYS

BASIC CONCEPT		GASES		LIQUIDS & SOLIDS		ATOMIC STRUCTURE		CHEMICAL BONDING		THERMOCHEMISTRY		CHEMICAL EQUILIBRIUM		SOLUTIONS		ELECTRO CHEMISTRY		REACTION KINETICS		MACROMOLECULES		CHEM INDUS	
1	B	1	A	1	D	1	C	1	A	1	A	1	A	1	B	1	A	1	A	1	C	1	B
2	B	2	B	2	A	2	B	2	C	2	C	2	A	2	D	2	C	2	C	2	C	2	C
3	B	3	C	3	A	3	D	3	A	3	A	3	B	3	A	3	A	3	B	3	C	3	B
4	B	4	A	4	C	4	C	4	C	4	B	4	D	4	A	4	C	4	D	4	B	4	C
5	B	5	A	5	A	5	A	5	B	5	B	5	A	5	C	5	B	5	A	5	A	5	C
6	B	6	A	6	C	6	A	6	A	6	B	6	A	6	A	6	A	6	C	6	A	6	B
7	C	7	C	7	B	7	B	7	C	7	D	7	A	7	D	7	C	7	B	7	B	7	A
8	B	8	A	8	A	8	A	8	A	8	A	8	B	8	C	8	A	8	A	8	A	8	C
9	D	9	C	9	C	9	B	9	D	9	A	9	B	9	D	9	A	9	D	9	C	9	A
10	D	10	A	10	D	10	C	10	C	10	B	10	A	10	B	10	B	10	D	10	A	10	B
11	A	11	A	11	B	11	C	11	B	11	C	11	C	11	A	11	A	11	C	11	B	11	D
12	D	12	C	12	C	12	C	12	A	12	D	12	C	12	D	12	C	12	C	12	C	12	B
13	B	13	A	13	C	13	D	13	D	13	C	13	A	13	C	13	A	13	A	13	D	13	B
14	C	14	C	14	C	14	A	14	B	14	A	14	A	14	C	14	A	14	A	14	C	14	C
15	D	15	C	15	D	15	A	15	C	15	C	15	C	15	B	15	A	15	A	15	C	15	A
16	B	16	A	16	D	16	D	16	C	16	C	16	A	16	D	16	B	16	C	16	A	16	D
17	C	17	C	17	C	17	B	17	A	17	B	17	A	17	B	17	A	17	A	17	D	17	D
18	B	18	C	18	B	18	B	18	C	18	A	18	D	18	B	18	C	18	C	18	B	18	A
19	B	19	D	19	A	19	B	19	A	19	C	19	A	19	D	19	B	19	B	19	B	19	D
20	B	20	C	20	B	20	B	20	D	20	D	20	C	20	C	20	B	20	D	20	A	20	C
21	D	21	B	21	C	21	A	21	C	21	A	21	D	21	A	21	B	21	C	21	C	21	C
22	C	22	B	22	C	22	A	22	B	22	C	22	D	22	C	22	A	22	C	22	A	22	A
23	D	23	D	23	A	23	C	23	C	23	C	23	C	23	D	23	B	23	C	23	D	23	C
24	D	24	A	24	D	24	C	24	A	24	D	24	A	24	B	24	C	24	A	24	B	24	
25	B	25	B	25	B	25	B	25	C	25	A	25	B	25	A	25	B	25	C	25	C	25	
26	B	26	C	26	B	26	A	26	A	26	B	26	B	26	C	26	A	26	A	26	A	26	
27	C	27	C	27	A	27	B	27	B	27	A	27	C	27	C	27	A	27	D	27	A	27	
28	A	28	D	28	C	28	B	28	B	28	C	28	B	28	C	28	B	28	D	28	D	28	
29	C	29	A	29	D	29	C	29	A	29	A	29	A	29	A	29	A	29	C	29	A	29	
30	B	30	B	30	D	30	C	30	C	30	C	30	A	30	C	30	B	30	B	30	B	30	
31	A	31	C	31	C	31	B	31	B	31	C	31	C	31	B	31	C	31	C	31	A	31	
32	B	32	D	32	A	32	A	32	A	32	A	32	A	32	B	32	A	32	B	32	D	32	
33	B	33	B	33	B	33	B	33	D	33	B	33	C	33	A	33	A	33	A	33	B	33	
34	A	34	A	34	B	34	B	34	A	34	C	34	C	34	B	34	B	34	D	34	C	34	
35	A	35	B	35	C	35	C	35	C	35	A	35	A	35	B	35	B	35	A	35	C	35	
36	B	36	B	36	B	36	A	36	A	36	B	36	B	36	D	36	D	36	B	36	D	36	
37	B	37	C	37	D	37	B	37	D	37	C	37		37	D	37	B	37	B	37	C	37	
38	A	38	B	38	D	38	C	38	C	38	D	38		38	A	38	B	38	B	38	D	38	
39	D	39		39	D	39	C	39	D	39	C	39		39		39	C	39	D	39	C	39	
40	B	40		40		40	C	40		40	D	40		40		40		40	D	40	B	40	
																				41	C		

ANSWER KEYS

PERIODIC CLASSIFICATION		S-BLOCK ELEMENTS		GROUP IIIA & IVA ELEMENTS		GROUP V-A VI-A ELEMENTS		HALOGENS & NOBLE GAS		TRANSITION ELEMENTS		FUNDAMENTAL OF ORGANIC CHEMISTRY		ALIPHATIC HYDROCARBONS		AROMATIC HYDROCARBONS		ALKYL HALIDES		ALCOHOL PHENOL ETHER		EXPERIMENTAL TECHNIQUES	
1	D	1	B	1	D	1	C	1	B	1	C	1	C	1	A	1	D	1	D	1	C	1	B
2	D	2	B	2	A	2	C	2	C	2	C	2	A	2	C	2	D	2	B	2	C	2	C
3	D	3	A	3	D	3	B	3	C	3	C	3	B	3	C	3	C	3	B	3	A	3	A
4	B	4	B	4	B	4	B	4	B	4	B	4	A	4	B	4	C	4	A	4	A	4	D
5	D	5	D	5	C	5	D	5	C	5	B	5	A	5	D	5	B	5	A	5	C	5	C
6	C	6	A	6	B	6	D	6	D	6	A	6	C	6	B	6	A	6	A	6	B	6	C
7	B	7	D	7	C	7	C	7	B	7	C	7	D	7	A	7	C	7	C	7	A	7	A
8	B	8	C	8	B	8	C	8	D	8	B	8	B	8	D	8	A	8	A	8	D	8	A
9	A	9	A	9	D	9	A	9	B	9	B	9	A	9	D	9	A	9	B	9	C	9	A
10	A	10	A	10	C	10	B	10	A	10	A	10	A	10	C	10	A	10	C	10	B	10	C
11	C	11	C	11	D	11	B	11	D	11	C	11	B	11	A	11	B	11	B	11	A	11	C
12	B	12	B	12	B	12	B	12	A	12	B	12	C	12	D	12	B	12	A	12	A	12	D
13	D	13	D	13	D	13	A	13	A	13	B	13	B	13	D	13	C	13	C	13	B	13	C
14	B	14	A	14	D	14	C	14	C	14	B	14	A	14	D	14	A	14	B	14	A	14	C
15	A	15	C	15	C	15	D	15	B	15	B	15	B	15	A	15	D	15	B	15	D	15	A
16	D	16	B	16	C	16	B	16	C	16	A	16	B	16	A	16	C	16	B	16	A	16	
17	B	17	A	17	D	17	D	17	A	17	C	17	D	17	A	17	D	17	C	17	D	17	
18	B	18	B	18	D	18	A	18	D	18	C	18	A	18	A	18	D	18	C	18	B	18	
19	C	19	A	19	C	19	C	19	D	19	C	19	A	19	A	19	B	19	B	19	C	19	
20	C	20	C	20	B	20	C	20	D	20	B	20	C	20	D	20	C	20	C	20	B	20	
21	D	21	A	21	C	21	B	21	C	21	A	21	B	21	A	21	D	21	A	21	A	21	
22	A	22	C	22	D	22	A	22	C	22	A	22	D	22	C	22	B	22	C	22	B	22	
23	A	23	A	23	B	23	D	23	D	23	A	23	B	23	B	23	B	23	A	23	A	23	
24	C	24	C	24	C	24	D	24	C	24	A	24	A	24	A	24	B	24	D	24	D	24	
25	D	25	D	25	B	25	A	25	C	25	A	25	D	25	B	25	A	25	C	25	A	25	
26	B	26	D	26	D	26	A	26	A	26	B	26	B	26	A	26	D	26	C	26	B	26	
27	D	27	D	27	C	27	A	27	D	27	B	27	A	27	D	27	D	27	B	27	C	27	
28	B	28	D	28	B	28	A	28	A	28	D	28	A	28	C	28	D	28	C	28	B	28	
29	D	29	D	29	A	29	C	29	A	29	B	29	D	29	C	29	A	29	B	29	C	29	
30	D	30	A	30	B	30	A	30	A	30	C	30	A	30	C	30	D	30	B	30	B	30	
31	A	31	B	31	B	31	D	31	B	31	B	31	A	31	C	31		31	C	31	D	31	
32	D	32	A	32	C	32	B	32	A	32	D	32	D	32	C	32		32	A	32	A	32	
33	A	33	C	33	C	33	C	33	A	33	D	33	D	33	B	33		33	D	33	A	33	
34	A	34	D	34	B	34	D	34	D	34	D	34	B	34	D	34		34	D	34	C	34	
35		35	B	35	A	35	D	35	D	35	D	35	A	35	C	35		35	D	35	C	35	
36		36	A	36	B	36	B	36	A	36	D	36	B	36	B	36		36	C	36	A	36	
37		37	B	37	D	37	C	37	C	37	A	37	C	37	A	37		37	D	37	C	37	
38		38	A	38	C	38	C	38	D	38	C	38	A	38	C	38		38	B	38	D	38	
39		39	C	39	C	39	B	39	C	39	A	39	D	39	B	39		39	A	39		39	
40		40	B	40		40	A	40	D	40		40	C	40		40		40	B	40		40	